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Software Requirements Specification

v1.0

Canari
For Playbook

Document Control:

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Disclaimer:

All the attached snapshots serve as reference for the fields, parameters, and functionalities of the platform's website, modules/sub-modules/sections. It may change at the time of actual development.

Table of Contents

1. Introduction.....	7
1.1. Document Purpose	7
1.2. Project Purpose	7
1.3. Project Objectives.....	7
1.4. Project Stakeholders	7
1.5. User Personas.....	8
1.5.1 Admin	8
1.5.2 Customer	8
1.5.3 Caller.....	8
1.6. Intended Audience	8
1.7. Acronyms and Abbreviation	8
1.8. Assumptions	9
1.9. Dependencies.....	10
1.10. System Security	10
1.11. System Feature	11
1.12. External Hardware	14
1.13. External Software	14
1.14. External System	15
2. Scope	15
3. Functional Requirement – General	20
4. Functional Requirement – Website.....	21
5. Functional Requirement – Admin.....	22
5.1 Login	22
5.2 Dashboard	23
5.3 Customers	25
5.4 Subscriptions	32
5.5 Ask Canari.....	35
5.6 Support.....	37
5.7 Settings.....	38
5.7.1 Account.....	38
5.7.2 Roles	39
5.7.3 Users.....	41

5.7.4	Calendars	42
5.7.5	Billing	44
5.7.6	Help and Resources	45
6.	Functional Requirement – Customer	47
6.1	Login	47
6.2	Dashboard	48
6.3	Campaigns	51
6.4	Events	54
6.5	Ask Canari	56
6.6	Help and Resources	57
6.7	Settings	58
6.7.1	Account	58
6.7.2	Users	60
6.7.3	Outlets	61
6.7.4	Groups	64
6.7.5	Canari Numbers	66
6.7.6	Billing	67
7.	Non-Functional Requirement	68
7.1	Performance	68
7.2	Scalability	69
7.3	Reliability	69
7.4	Usability	69
7.5	Security	69
7.6	Maintainability	69
7.7	Compatibility	69
8.	System Requirement	69
8.1	User Interface	69
8.2	Database	70
8.3	Communication/Security	70
8.4	System Integration	70
8.5	Constraints and Limitations	71
8.6	Legal	71
8.6.1	Data Protection and Privacy	71

8.6.2	Call Recording Regulations	71
8.6.3	Liability Limitation	71
9.	System Architecture	71
9.1	Overview	71
9.2	N-tier Architecture	72
9.2.1	Presentation Component	72
9.2.2	Business Logic Component	72
9.2.3	Data Access Component.....	73
9.2.4	Communication/Security Component.....	73
9.2.5	API Component.....	73
9.3	Microservices Architecture	74
9.3.1	Key Benefits	75
9.3.2	Core Microservices and Their Responsibilities	75
9.3.3	Communication and Integration Patterns	77
9.4	Method of Development	78
9.5	High-Level Technical Design	79
10.	Acceptance Criteria	79
11.	Product Future.....	80
12.	Technical Evaluation of Call Providers.....	81
12.1	Functionalities	81
12.2	Pricing.....	81
13.	AI Requirement	82
13.1	Intent Detection	82
13.2	Emotion Recognition	82
13.3	Call Reason Categorisation	83
13.4	Customer Satisfaction Prediction	83
13.5	Average Call Duration.....	83
13.6	Sentiment Trajectory.....	84
13.7	Call Transcription.....	84
13.8	Time Series and Trend Analysis	84
13.9	Gender Detection	85
13.10	Entity Extraction	85
13.11	Sentiment Analysis	85

13.12	Demographic Inference - Gender, Tourist vs Local.....	86
13.13	Ask Canari	86
13.14	New vs Returning Customer - Phrases Said and Recognising Phone Number.....	87
13.15	Lost Opportunities	87
13.16	Complaints Section	87
13.17	Peak Call Times & Projections	88
14.	Integration of Third-Party Data: Holiday Calendars by Country.....	88
14.1	Integration Approaches.....	88
14.2	Admin Portal Integration.....	89
14.3	Benefits.....	89
14.4	Implementation Considerations	89
14.5	Model Flexibility and Adaptability.....	90
14.6	Conclusion	90
15.	Stripe Integration and Features.....	90
15.1	Key Objectives	90
15.2	Stripe Features Utilised	90
15.2.1	Payment Processing.....	90
15.2.2	Subscription Management	91
15.2.3	Invoicing	91
15.2.4	Webhooks.....	91
15.2.5	Fraud Detection and Prevention	91
15.2.6	PCI DSS Compliance	91
15.2.7	Financial Reporting and Analytics.....	91
15.2.8	Customer Portal.....	91
15.3	Integration Strategy	91
15.4	Fee Structure (UK)	92
15.5	Security and Compliance	93
15.6	Workflow	94
16.	CMS – Webflow	95
17.	Infrastructure Requirement	96
17.1	Scenario	96

1. Introduction

1.1. Document Purpose

The purpose of this document is to serve as a comprehensive guide for the development and implementation of the **Canari** platform. The document aims to ensure all stakeholders have a unified understanding of the system's capabilities and how it will support the intended business processes. It acts as a reference point throughout the project lifecycle, ensuring that development aligns with customer expectations and industry best practices. Furthermore, it facilitates effective communication between technical teams and business stakeholders, promoting transparency and reducing the likelihood of misinterpretations during the development phase.

1.2. Project Purpose

The purpose of this project is to address the growing need for hospitality brands to make data-driven decisions in their marketing efforts. As customer engagement becomes increasingly competitive, there is a critical need to understand and leverage inbound call activities to optimise marketing campaigns and enhance customer satisfaction. This platform is designed to provide actionable insights from call analytics, helping businesses improve customer interactions, streamline operations, and increase ROI. This platform enables hospitality brands in the UK, USA and UAE to remain competitive and meet their evolving business needs.

1.3. Project Objectives

Project objectives establish criteria for the success of the project, and outline specific benchmarks to be achieved during its execution:

- **Build a Scalable SaaS Platform:** Develop a flexible and robust cloud-based SaaS platform capable of supporting the growing needs of hospitality brands across the UK, USA and UAE.
- **Provide Actionable Insights:** Implement advanced features (powered by AI), including call analytics, transcription, and sentiment analysis, to convert raw data into meaningful insights for improved decision-making.
- **Enhance Customer Engagement:** Equip businesses with tools to monitor and enhance customer interactions, ensuring higher satisfaction and increased loyalty.
- **Improve Marketing Effectiveness:** Enable brands to fine-tune their marketing strategies by providing detailed insights into campaign performance based on call analytics.
- **Ensure Ease of Use:** Design a user-friendly interface that allows customers to effortlessly navigate the platform and utilise its features without technical expertise.
- **Support Long-Term Growth:** Develop the platform with scalability in mind, ensuring it can adapt to evolving industry needs and expanding business operations.

1.4. Project Stakeholders

Role	Name	Email ID
Client		
Founder	Josh Ashley-Collins	Josh@theplaybookme.com
Director	Sadie Kehoe	sadie@theplaybookme.com
Sales Head	Tim Howell-Ives	tim@theplaybookme.com
Marketing Executive	Natan Kupczyk	natan@theplaybookme.com

MSBC		
COO	Rishan Gopal	rishan.gopal@msbcgroup.com
Lead AI Engineer	Suhail Ahmed	suhail.ahmed@msbcgroup.com
Sr. UI - UX Designer	Aesha Thakkar	Aesha.Thakkar@msbcgroup.com
Partnerships Manager	Shubhi Mittal	shubhi.mittal@msbcgroup.com
Business Analyst	Ajay Chakrobarty	ajay.chakrobarty@msbcgroup.com

1.5. User Personas

1.5.1 Admin

The **Admin** is the owner and primary manager of this SaaS (**Canari**) platform. This user is responsible for overseeing the entire system, managing customer registrations, handling subscription plans, etc. Admin users have access to the Admin Portal, where they can perform critical tasks, such as taking decisions based on the call analytics, adding or updating customers, managing user permissions, monitoring platform usage, etc. The Admin also provides support to customers, ensuring they have the necessary resources and guidance to utilise the customer portal effectively.

1.5.2 Customer

The **Customer** represents businesses or individuals who have subscribed to the **Canari** platform to utilise its call insights and data analytics features. Customers have access to the Customer Portal, where they can manage their accounts, campaigns, inbound call analytics etc. They can make data-driven decisions by leveraging the platform's insights to improve their marketing campaigns and overall customer engagement. Customers also can manage their teams by adding users to collaborate within the platform.

1.5.3 Caller

A **Caller** is an external user who interacts with the platform by calling canari phone numbers associated with the customer's business, typically for inquiries, support, or product/service-related issues. Although Callers do not have direct access to the platform, their interactions are tracked and analysed to provide valuable insights for both the admin and customer. The platform captures key metrics such as booking value, inquiries value, the reason for the call, the number of missed calls along with potential revenue loss, call duration, call outcomes, sentiment analysis etc. to help both admins and customers make informed decisions.

1.6. Intended Audience

The **Canari** platform is specifically designed for hospitality brands operating in the **UK** (this would be the **primary market** in the initial phase) with plans of catering to brands operating in other countries in the future. The platform targets restaurants, hotel chains, and other hospitality businesses that rely heavily on customer engagement and marketing campaigns to drive business success. By providing actionable insights through call analytics, this platform enables hospitality brands to optimise their customer interactions, improve marketing effectiveness, and increase overall **ROI**.

1.7. Acronyms and Abbreviation

Key	Full Form
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ROI	Return on Investment
SaaS	Software as a Service
API	Application Programming Interface
EULA	End User Licence Agreement
FAQ	Frequently Asked Questions
IVR	Interactive Voice Response
NLP	Natural Language Processing
AWS	Amazon Web Services
RBAC	Role Based Access Control
TLS	Transport Layer Security
SSL	Secure Sockets Layer
SQL	Structured Query Language
XSS	Cross Site Scripting
CSRF	Cross Site Request Forgery
SDLC	Software Development Life Cycle
STAMP	Key Performance Indicator
CRM	Customer Relationship Management
POS	Point of Sale
SDK	Software Development Kit
HTTPS	Hypertext Transfer Protocol Secure
WAF	Web Application Firewall
CA	Certificate Authorities
CDN	Content Delivery Network
GDPR	General Data Protection Regulation
GCP	Google Cloud Platform
AO	Account Owner
FOC	Free of Cost
ACID	Atomicity, Consistency, Isolation, and Durability
DBMS	Database Management System
ORM	Object Relational Mapping
IAM	Identity and Access Management
CI/CD	Continuous Integration and Continuous Delivery/Deployment

1.8. Assumptions

- **Language Support:** The platform will initially support only the **English** language.
- **Customer Infrastructure:** Customers possess the necessary hardware and stable internet connectivity to access and use the platform effectively.
- **Call Provider Integration:** The availability and reliability of third-party call providers offering APIs for call connection, recording, transcription, and other call management functionalities are assumed.
- **Legal Documentation:** The canari team will provide the necessary legal documents, including the End-User Licence Agreement (EULA), Privacy Policy, and Terms & Conditions.
- **Data Security:** The platform will not store any sensitive payment information such as credit or debit card details.

- **Customer Location:** Initially, all customer outlets will be exclusively located within the United Kingdom.
- **User Registration:** Customers will provide valid and accurate information during registration, including a functional email address for communication and account setup.
- **Customer Feedback:** Customers will provide feedback to support the development process.
- **Data Retention Policy:** The platform will retain call data and analytics for a predefined period (e.g., 2-3 months after account cancellation) unless otherwise required by law.
- **Required Customer Information:** The information asked and/or required by the Canari team from the respective customer is subject to change in case the call provider changes the requirement from their end.
- **User Training and Support:** Assumes that customers will utilise provided resources such as video tutorials, FAQs, and support chat to familiarise themselves with the platform.

1.9. Dependencies

- **Calendly:** For scheduling demos and meetings with potential customers.
- **Stripe:** For secure payment processing and subscription management.
- **Call Providers:** Integration with a third-party call provider (**Telnyx**) for handling calls, recordings, and transcriptions.
- **AI and NLP Services:** Utilising external AI services or APIs for features like call transcription, sentiment analysis, keyword extraction, and emotion detection.
- **Google Analytics:** For tracking website traffic data and user engagement metrics.
- **Internet Connectivity:** Stable and reliable internet connections are essential for both the platform's servers and the users accessing the service.
- **Cloud Infrastructure:** Dependence on cloud services for data storage, processing, and scalability (e.g., AWS).
- **Regulatory Environment:** Compliance with regional data protection and privacy laws is mandatory. Changes in regulations may require platform adjustments.
- **Notification Services:** Dependence on email and notification services for sending automated emails, reminders, and alerts to users.

1.10. System Security

- **Authentication and Authorisation:** Utilise **RBAC** to ensure users have access only to the functionalities and data pertinent to their roles, minimising the risk of unauthorised access.
- **Data Encryption:**
 - **Encryption at Rest:** All sensitive data stored within the platform will be encrypted using industry-standard encryption algorithms (e.g., **AES-256**) to protect data even if storage media are compromised.
 - **Encryption in Transit:** Utilise TLS/SSL protocols to encrypt data transmitted between the customer, server, and third-party services, preventing interception and eavesdropping.
- **Secure Coding Practices:**
 - **Code Reviews:** Regular code reviews and security audits will be conducted to identify and remediate vulnerabilities such as SQL injection, cross-site scripting (XSS), and cross-site request forgery (CSRF).

- **Secure Development Lifecycle:** Adopt a secure **SDLC** that integrates security testing at every phase of development.
- **Network and Infrastructure:**
 - **Security Firewalls and Intrusion Detection Systems:** Deploy firewalls and intrusion detection/prevention systems to monitor and protect the network infrastructure from malicious activities.
 - **Server Hardening:** Implement server hardening techniques, including disabling unnecessary services and ports, to reduce the attack surface.
- **Security Monitoring and Incident Response:**
 - **Real-Time Monitoring:** Implement real-time security monitoring tools to detect suspicious activities promptly.
 - **Incident Response Plan:** Develop and maintain an incident response plan outlining procedures for addressing security breaches, including communication protocols and mitigation strategies.
- **Backup and Disaster Recovery:**
 - **Data Backups:** Implement regular automated backups of critical data to secure, encrypted storage locations.
 - **Disaster Recovery Plan:** Establish a disaster recovery plan to restore services rapidly in the event of a catastrophic failure or data loss incident.
- **Third-Party Security Management:**
 - **Vendor Assessments:** Evaluate third-party service providers for their security practices to ensure they meet the platform's security requirements.
 - **Data Handling Agreements:** Establish clear agreements with third-party providers regarding data protection responsibilities and breach notification procedures.
- **Secure APIs:**
 - **API Security:** Protect APIs used for integration with external services through server validation.
 - **API Key Management:** Securely manage API keys and credentials, ensuring they are stored encrypted and rotated periodically.
- **Data Anonymisation and Minimisation:**
 - **Anonymisation Techniques:** Employ data anonymisation where applicable to protect personal information within analytics and reports.
 - **Data Minimisation:** Collect only the necessary data required for platform functionality to reduce exposure of sensitive information.

1.11. System Feature

- **Advanced Analytics Dashboard (Powered by AI):**
 - **Interactive Visualisations:** Analytics and insights that display key metrics such as upcoming bookings along with values, average response time, reason for inquiries, missed calls along with potential values, peak call times, average call duration, and customer sentiment over time.
 - **Customisable Views:** Users can filter data by date range, campaign, event or specific keywords to focus on areas of interest.
 - **Widgets:** Widgets are interactive data components displayed on various screens, designed to provide an overview of information in a compact and visually accessible format. Widgets are expandable, meaning that users can click on a widget to view more detailed information or additional insights related to the displayed data. This

expandability offers a deeper, more interactive experience, allowing users to engage with and explore the data in greater detail.

- **Stamps:** Stamps represent specific, measurable metrics that provide insights into the performance of various business objectives. These indicators are displayed on multiple screens within the application to provide users with at-a-glance data relevant to their business goals. Unlike widgets, Stamps are typically static and do not feature expandability. Their primary function is to present concise, real-time data without additional interaction or detailed views.
- **Call Recording and Transcription:**
 - **Automatic Call Recording:** Securely records inbound calls for analysis and quality assurance.
 - **High-Accuracy Transcriptions:** Converts voice calls into text using advanced speech-to-text technology.
- **Sentiment and Emotion Analysis:**
 - **AI-Powered Sentiment Detection:** Analyses transcribed calls to determine the caller's sentiment (positive, neutral, negative).
 - **Emotion Recognition:** Identifies emotions such as happiness, frustration, or urgency to better understand customer needs.
 - **Sentiment Trajectory Analysis:** Tracks how the sentiment changes throughout the call for deeper insights into customer experience.
- **Keyword, Topic, and Entity Extraction:**
 - **Frequent Keywords Identification:** Highlights commonly used words or phrases to reveal trends or common inquiries.
 - **Topic Clustering:** Groups related keywords to identify overarching themes in customer conversations.
 - **Entity Extraction:** Extracts key entities like names, dates, menu items, or locations from the conversation for actionable insights.
- **Intent Detection and Call Reason Categorisation:**
 - **Intent Detection:** Identifies the purpose of the call, such as making a reservation, placing an order, or lodging a complaint.
 - **Call Reason Categorisation:** Classifies calls into predefined categories to streamline response strategies and staff training.
- **Customer Satisfaction Prediction:**
 - **Satisfaction Scoring:** Predicts customer satisfaction levels based on call transcript and tone using machine learning models.
- **Campaign and Phone Number Management:**
 - **Unique Tracking Numbers:** Assigns specific phone numbers to different marketing campaigns for precise attribution.
 - **Number Provisioning and Porting:** Allows customers to acquire new numbers or bring existing ones into the platform.
- **User Management and Roles:**
 - **Role-Based Access Control:** Defines user permissions (e.g., Admin, Manager, Analyst) to ensure appropriate access levels.

- **Multi-User Support:** Enables collaboration within teams by allowing multiple users under a single customer account.
- **Automated Alerts and Notifications:**
 - **Customisable Triggers:** Sends alerts for events like missed calls.
 - **Notification Channels:** Delivers alerts via email and/or within the platform's notification centre (depending upon scenarios).
- **Data Export and Reporting:**
 - **Report Generation:** Export detailed reports of each STAMP/screen in two formats: PDF and Excel.
 - **Scheduled Reports:** Automates the delivery of reports at regular intervals to designated recipients.
- **Integration Capabilities:**
 - **API Access:** Offers APIs for customers who wish to integrate call data with their existing systems.
 - **Future Integration Plans:** Lays the groundwork for integrating with CRM, POS, and booking systems in subsequent development phases.
- **Support and Educational Resources:**
 - **In-App Guidance:** Tooltips and onboarding wizards to help users navigate the platform.
 - **Resource Library:** Access to FAQs, video tutorials, and documentation for self-help.
- **Components:**
 - **Core:** Core components are those that are essential to the business objectives of the customer. These components are refreshed on a real-time basis to ensure up-to-date, actionable insights that are available for decision-making. All the core components will be indicated with a **Green** circle. The following are the components that are continuously refreshed, granted there is new data to be computed.
 - Missed Calls
 - Approximate Loss
 - Average Response Time
 - Time Taken to Answer a Call
 - Minutes Remaining
 - Peak Call Time (Not Future)
 - Lifetime Insights
 - **Non-core:** Non-core components are those elements within the application that do not require immediate updates and are computationally intensive. These components are refreshed on a periodic bases and provides a balanced and insightful view of the data, offering meaningful updates while minimising the load on the system. The following are the Non-core components that are continuously refreshed once the respective data are available:
 - Upcoming Bookings
 - Total Inquiries
 - Lost Opportunities
 - Why People Called
 - Complaints Section

- Future Peak Call Times
- Local vs. Tourist
- Gender
- Satisfaction Rate
- New vs. Returning Customers
- Campaign Tracking
- Event Tracking

1.12. External Hardware

- **Customers and Administrators:** Require standard desktop or laptop computers capable of running modern web browsers efficiently. These devices should meet minimum specifications, such as:
 - **Processor:** Dual-core CPU at 2 GHz or higher
 - **Memory:** 4 GB RAM or more
 - **Display:** Resolution of 1280x720 pixels or higher
 - **Internet Connectivity:** Stable broadband connection with at least 5 Mbps download/upload speeds
- **Callers:**
 - **Telephone Devices:** Callers use standard telephones or mobile devices to place calls to the customers' business numbers. No special hardware is required from callers.
 - **Server Infrastructure:** The platform relies on external cloud service providers like AWS for server hardware, storage, and network infrastructure. This includes high-performance servers, secure data storage solutions, and reliable networking equipment. The management and maintenance of this hardware are the responsibility of the cloud providers (e.g., AWS, Azure).

1.13. External Software

- **Supported Browsers:** The latest versions of Google Chrome, Mozilla Firefox, Microsoft Edge, and Apple Safari are recommended for optimal performance and compatibility.
- **Third-Party APIs and SDKs:**
 - **Call Management APIs:** Integration with APIs from third-party call providers (**Telnyx**) to handle call connection, recording, transcription, and analytics.
 - **Payment Processing Software:** Utilisation of **Stripe's** SDKs and APIs for secure handling of subscription payments and billing management.
 - **Scheduling Tools:** Integration with **Calendly** for seamless scheduling of demos and meetings.
- **AI Analytics:** Integration with AI tools (if needed) such as OpenAI GPT-4, IBM Watson Tone Analyser, Azure Machine Learning, Amazon Forecast, Hugging Face etc. to analyse the raw data and provide actionable insights.
- **Analytics and Monitoring Tools:**
 - **Google Analytics:** For tracking website traffic, user behaviour, and engagement metrics.
 - **Server Monitoring Software:** Tools like **New Relic** or **Datadog** for monitoring server performance, uptime, and resource utilisation.

- **Email Services:** Services such as **SendGrid** or **Mailgun** are used for sending automated emails (e.g., account confirmations, password resets, notifications etc.).
- **Security Software:**
 - **SSL/TLS Certificates:** Provided by trusted Certificate Authorities (CAs) to secure data transmission over HTTPS.
 - **WAF:** Software solutions that protect the platform from common web exploits and attacks.
- **CDN Services:** Utilisation of CDNs like **Cloudflare** or **Amazon CloudFront** to distribute static content efficiently to users worldwide, improving load times and performance.
- **GDPR Compliance Software:** Tools that help manage user consent, data requests, and other compliance-related activities.

1.14. External System

- **Telephony and Call Services:**
 - **Call Connection and Recording:** Managing incoming calls, directing them to the appropriate recipients, and recording them.
 - **Number Provisioning:** Providing phone numbers for campaigns and allowing number porting.
 - **APIs for Integration:** Enabling seamless integration of telephony functions into the platform.
- **Stripe:**
 - **Payment Processing:** Handling credit/debit card payments securely.
 - **Subscription Management:** Managing recurring billing cycles and invoicing.
 - **Fraud Detection:** Built-in tools to detect and prevent fraudulent transactions.
- **Calendly:**
 - **Calendar Integration:** Syncing with Google Calendar, Outlook, and others.
 - **Automated Reminders:** Sending email notifications to reduce no-shows.
- **Hosting and Infrastructure:**
 - AWS, Microsoft Azure, or GCP providing scalable and reliable cloud infrastructure for hosting the platform, including:
 - **Compute Services:** Virtual servers for running the application backend.
 - **Storage Services:** Secure data storage solutions for databases and files.
 - **Networking Services:** Virtual networks, load balancers, and security groups.
- **Email Delivery System:**
 - **SendGrid, Mailgun, or Amazon SES:** Used for reliable delivery of transactional emails, such as Welcome Emails, Password Reset Instructions and Notification Alerts.
- **CDN:** Cloudflare, Akamai or Amazon CloudFront CDNs are used to distribute static content globally, ensuring fast load times regardless of user location.

2. Scope

Sr. No	Module	Section	Features
Website			
1	Website	NA	1. Hero Section

			2. Features 3. Process 4. Introduction Video 5. Analytics 6. Client Testimonials 7. Calendly for Book a Demo
Admin Portal			
1	Dashboard	NA	1. AI summary of all the analytics of the home screen 2. Widgets (once selected any customer/group): Upcoming booking along with values, average response time, and missed calls along with potential revenue loss 3. Lost Opportunities, Call Reason, Complaints Section, Peak Call Hours (along with call volumes), satisfaction rate, Local vs Tourist, New vs Returning and Gender 4. Each widget/STAMP is expandable and once clicked showcases detailed information including call transcription (where applicable) 5. Export (PDF-friendly snap of the dashboard)
2	Customers	NA	1. Add customer and share stripe link 2. Details Screen: Outlets, Users, Dashboard, Campaigns, Canari Numbers, Ask Canari, Billing and Call Transcription 3. Beta functionality, group creation (Admin) 4. As per the customer's request, the admin can add Outlet, User, Campaign, and Event on behalf of the respective customer
3	Subscriptions	NA	1. Master lists of Subscription Plans (Monthly and Annually), Additional Number Plans and Additional Minutes Plans 2. Admin needs to manually enter the amount of the subscription plans 3. All the plans would be in \$ 4. The total allocated minutes for each customer will be assigned upfront, based on the selected subscription plan
4	Ask Canari	NA	1. Can ask questions related to both portals 2. Can see Customer's interaction 3. Only text-to-text interaction 4. Read Aloud functionality 5. Regenerate the result and submit the feedback

			6. Upon refreshing the page, the admin will no longer have access to their previous chat history
5	Support	NA	1. View all the tickets raised by all the customers and act accordingly 2. Can see all the submitted Features Request forms
6	Settings	Account	1. View and manage personal information
		Roles	1. Create, edit and manage user roles 2. Two types of permissions: View and Add (both will have download and export functionalities) 3. Minimum 1 view access needs to be enabled to save a record 4. Add access will automatically enable the View Access
		Users	1. Create, edit and manage users of the admin portal 2. A Sysadmin user will be added from the backend with admin role rights and cannot be editable
		Calendar	1. Calendar view 2. Default Current Month 3. Integrated API for holidays/occasions 4. Manually mark holidays/occasions 5. Auto Refresh (on every Monday) 6. Manual Refresh Button 7. Date Range Selection 8. Multi-Region Support 9. Future Dates Only 10. Holidays/Occasions Filtering/Search 11. Details Pop-Up 12. Regional Outlet Linking and call analytics 13. Read-Only Past Records
		Billing	1. View all the billing details of all the customers 2. Widgets + Listing Screen 3. Upon clicking the View button, the admin will be redirected to the Stripe page, where the system will display additional details for the selected invoice 4. Can download the invoice
		Help and Resources	1. Create and manage content for both portals and website 2. Images and Videos could be uploaded along with relevant Thumbnail
Customer Portal			

1	Dashboard	NA	1. AI summary (Text format) of all the analytics of the home screen 2. Widgets: Upcoming booking along with values, average response time, and missed calls along with potential revenue loss 3. Lost Opportunities, Call Reason, Complaints Section, Peak Call Hours (along with call volumes), satisfaction rate, Local vs Tourist, New vs Returning and Gender 4. My Campaign: Up to 4 favourite/most engaged campaigns 5. My Events: Up to 4 favourite/most engaged events 6. Each widget/STAMP is expandable and once clicked showcases detailed information including 7. Export (PDF-friendly snap of the dashboard)
2	Campaigns	NA	1. Create and manage campaigns 2. Assign a phone number is must to create a campaign 3. Mark as Favourite (up to 4) 4. Additional Phone Numbers/Minutes purchase 5. Detailed view: All the listing screen's information along with widgets and AI summary (text format) 6. Export (PDF and Excel)
3	Events	NA	1. Create and manage events (group-wide promotion) 2. Assign groups or outlets is a must to create an event 3. Mark as Favourite (up to 4)
4	Ask Canari	NA	1. Can ask questions related to assigned outlet(s)/group(s) (Customer portal) 2. Only text-to-text interaction 3. Read Aloud functionality 4. Regenerate the result and submit the feedback 5. Upon refreshing the page, the customer will no longer have access to their previous chat history
5	Profile	Lifetime Insights	1. Visual representation of the customer's business growth trajectory since they signed up for the application 2. Provide insights into the positive trends and improvements in the customer's business performance, highlighting key milestones and overall savings achieved over time

			3. Download Option (as JPEG)
		Help and Resources	4. Access contents and raise a support ticket or feature request 5. All the contents will have a title and description along with relevant videos 6. Customer will get emails for support ticket
6	Settings	Account	1. View and manage personal information 2. Set up email frequency 3. Cancel subscription
		Users	1. Create, edit and manage users of the customer portal 2. Two types of permissions: View and Add (both will have download and export functionalities) 3. Minimum 1 view access needs to be enabled to save a record 4. Add access will automatically enable the View access
		Outlets	1. Set up outlets 2. Need to enter outlet setup questionnaire 3. Can redirect to the dashboard and see the information related to selected outlet only 4. Mark as Favourite (up to 5)
		Groups	1. Create and manage groups 2. Card format 3. A minimum of two outlets need to be selected to create a group 4. By default, only the account owner can create a group
		Canari Numbers	1. View and manage associate canari phone numbers 2. Widgets + Listing Screen 3. Can buy Additional Phone Number(s)/Minutes 4. By default, only the account owner can create a group
		Billings	1. View invoices 2. Listing Screen 3. Automated payment for services like subscription, additional canari number and minutes 4. Auto top-up once reached 90% 5. Upon clicking the View button, the customer will be redirected to the Stripe page, where the system will display additional details for the selected invoice 6. Can download the invoice

General			
1	Login	NA	1. Email Address + Password 2. Magic Link 3. Forgot Password
2	Compliance	Data Security & Privacy	1. Data-at-rest and data-in-transit encryption
		Regulatory & Regional Compliancy	1. GDPR compliance

3. Functional Requirement – General

- **User Role Management:** The access to both platforms (Admin and Customer) will be governed by the roles/permissions assigned to the logged-in user. This role-based access control will ensure that users can only access functionalities pertinent to their roles.
- **User Interface Elements:** The screen will display the logged-in user's name, the platform version, and copyright information. Additionally, tooltips, privacy settings and a collapsible navigation menu will be available to enhance user experience and accessibility. All the data would be bound with the default sort order of latest to oldest.
- **Call Connection Flow:** The system requires the customer to display the designated Canari numbers across all relevant platforms, including the website, social media, campaigns, etc. When a caller dials this specific Canari number, the system will connect the incoming call to the associated outlet's phone number. The system will record the call and generate a transcription. Based on the configured frequency (frequency set by the canari admin in the admin portal **for Non-Core components** and **instant** for **Core components**), the system will begin processing analytics for the recorded call. Once the analysis is complete, the system will display the relevant data and insights across both the Admin and Customer portals in the appropriate modules/sections.
- **Listing Screen:** All the listing screen would support the following features by default across the applications (Admin and Client Portal):
 - Sorting
 - Filtering
 - Export
 - Search
 - Pagination
- **Application Records Update History:** The platform will maintain an update history for all records, capturing the following details:
 - Field
 - Record ID
 - Updated Field
 - Updated From
 - Updated To
 - Updated By
 - Updated On
- **Last Updated On:** The system will display the "Last Updated On" timeStamp for both portals, reflecting the most recent refresh time of the Non-Core components. This timeStamp will provide users with visibility into when the data was last updated. However, for historical data, the system will not display a "Last Updated On" timeStamp, as it pertains to data that is not subject to real-time updates or refresh cycles.

- **User Session Management:** The logged-in user session will automatically expire after 30 minutes of inactivity to enhance security and resource management.
- **Device Compatibility:** Need to design a responsive and intuitive interface compatible with devices such as desktops, tablets, and mobiles for the **Website** and **Customer Portal**. The Admin portal would be a web application.
- **Default Currency:** The default currency unit for both the Admin and Customer portals will be USD (\$). Users will have the option to change the currency unit through the settings menu. A Currency API will execute once daily to update and adjust the currency exchange rates accordingly.
- **Phone Number Validity:** Once assigned, a phone number will be associated with the respective customer, even after the subscription ends. If the customer does not renew the subscription, the associated phone numbers will remain linked to the customer for a period of **three months**. After three months, if the subscription is not renewed, the system will release the phone numbers, making them available for reassignment to new customers as needed.
- **Remaining Minutes:** The system will display the remaining minutes on the customer portal for each respective customer, providing real-time visibility into their usage.
- **Lifetime Insights:** This functionality will be accessible to customers through the **Profile** section of the Customer Portal. Upon navigating to the Profile page, users will find a button labelled **Lifetime Insights**. When selected, the system will display a visual representation of the customer's business growth trajectory since they signed up for the application. This feature will provide insights into the positive trends and improvements in the customer's business performance, highlighting key milestones and overall savings achieved over time. The information will be presented in an intuitive, user-friendly format to ensure easy interpretation of the data. The customer can export the information as a PDF or Excel.
- **Date Format Standardisation:** The date format across the application will be standardised to "**DD MMM YYYY**" to ensure consistency and clarity.

4. Functional Requirement – Website

The website serves as a primary interface for customers to explore platform features, book a demo, and for existing users to log in to the customer portal.	
Preconditions	1) The customer must provide a valid email address. 2) Integration with Calendly should be active to enable demo booking. 3) Google Analytics tracking must be implemented to capture website traffic and user behaviour.
Post Conditions	1) The customer will successfully book a demo with the canari team. 2) Registered customers can log into the customer portal from the website. 3) The system tracks and logs all site traffic and interaction metrics via Google Analytics.
Normal Flows	1) The customer visits the Canari website. 2) The customer navigates to the Book a Demo section at the bottom of the page or clicks the Book a Demo button available in multiple places. 3) The system displays an embedded Calendly calendar. 4) The customer selects a date and time slot from the calendar and enters the following details: a) Name b) Email

	c) Notes 5) The customer clicks Submit. 6) The system books the selected demo timeslot and sends an automated invitation email to both the customer and the canari team with meeting details. 7) If the demo booking fails due to connectivity issues with Calendly, the system displays an error message and prompts the customer to try again or contact support. 8) If the customer submits incomplete information in the demo booking form, the system displays a prompt to fill in all required fields.
Field Validations	1) Name: Textbox, alphanumeric along with special character enabled, maximum length = 100 characters and mandatory. 2) Email: Textbox, alphanumeric along with special character enabled, format specific, maximum length = 250 characters and mandatory. 3) Notes: Text area, alphanumeric along with special character supported, maximum length = 1000 characters and optional.
Business Validations	1) All demo bookings must be recorded on Calendly and reflected in both the customer's and the canari team's calendars. 2) Google Analytics data should be accurate, capturing all essential metrics (e.g., page views, demo booking conversions) for insights and optimisation.
Notes	1) Ensure SSL certification for secure communication and data transmission on the website. 2) Regularly monitor the Google Analytics dashboard for site performance and user behaviour metrics to identify any usability improvements.

5. Functional Requirement – Admin

5.1 Login

This screen allows registered admins to securely access the Admin Portal by entering their registered email ID and password or by using the Magic Link .	
Preconditions	1) A Sysadmin user will be created in the admin portal from the backend for the canari. 2) The admin must listed in the Users screen with an Active status. 3) The admin must have completed their initial setup, including password creation.
Post Conditions	1) Admins are successfully authenticated and directed to the Dashboard. 2) The admin can initiate a password reset or Magic Link login if they are unable to remember their password.
Normal Flows	1) Upon launching the application, the admin will be prompted to access the login screen. 2) The admin enters their registered Email ID and Password into the respective fields. 3) The admin clicks the Sign In button.

	4) If the credentials match the system database, the admin is authenticated and directed to the Dashboard. 5) Forgot Password Flow: a) If the admin cannot remember their password, they click the Forgot Password button. b) The admin enters their registered email and clicks Submit. c) The system sends an email with a password reset link. d) Upon clicking the link, the admin is directed to a page to set and confirm a new password. 6) Magic Link Flow: a) The admin selects Get a Magic Link on the login page. b) The system sends an email with a one-time link to the admin's registered email. c) Upon clicking the link, the admin is redirected directly to the Dashboard without needing to enter a password.
Field Validations	1) Email Address: Textbox, alphanumeric, format specific (such as <u>user@domain.com</u>), mandatory and maximum length = 250 characters. 2) Password: Textbox, alphanumeric, format specific (must include minimum 1 letter, number and special characters), mandatory, minimum length = 8 characters and maximum length = 24 characters.
Business Validations	1) If the entered email or password does not match an existing record in the system, display an error message: The provided credentials are not registered with the application. 2) Automated email notifications should be sent for activities such as Forgot Password and Magic Link to ensure the security and traceability of login activities.
Notes	1) A rate limit should be set on unsuccessful login attempts to prevent brute-force attacks. 2) The Magic Link option should include an expiry time (e.g., 15 minutes) for added security.

5.2 Dashboard

This module provides an analytical overview of call data, campaign performance, and customer insights, enabling admins to monitor and evaluate all customer activities and performance metrics.	
Preconditions	1) The admin must be logged into the Admin Portal with permission to view the Dashboard. 2) The system must have access to customer data, including call analytics, campaign details, and Stamps. 3) Historical and real-time data must be available to populate Dashboard metrics.
Post Conditions	1) The admin can view, sort, and filter Stamps and widgets. 2) Detailed analytics, including call transcription, are available upon clicking a Stamp or widget.
Normal Flows	1) The admin logs into the Admin Portal, and the system redirects them to the Dashboard.

	2) The Dashboard displays the following sections (cumulative values of all the customers) in the first tab (All): a) First Section: An AI summary (text format) of all the analytics available on the home screen. b) Second Section (Widgets): Primary call analytics, such as upcoming booking along with values, average response time, and missed calls along with potential revenue loss. Each widget will display either an upward or downward arrow, accompanied by the corresponding percentage, based on the data for the selected date range. This section would not be visible by default and it will become visible only when the admin selects a specific customer or group (created by Admin) from the filter. c) Third Section: Lost Opportunities, call reason, complaints section, peak call hours (along with call volumes), satisfaction rate, local vs tourist, new vs returning, and gender. 3) The admin can use the search box to search for a specific customer or group (created by the admin). 4) The system will display all relevant search results. Upon selecting (single selection) a customer or group, the dashboard will update to display information based on the selected customer or group. 5) Once a customer or group is selected, the admin can then search for and view all associated outlets, groups (created by the customer), and campaigns linked to the initial selection. 6) The admin applies date filters to adjust the time range of displayed data. 7) By clicking on a Stamp or widget, the system displays detailed analytics such as age range, tourist vs local, and new vs returning callers, with associated call transcriptions. 8) The admin can use the export functionality to get the snap of the dashboard in a print-friendly PDF. The canari logo would be visible in that PDF. 9) While the widget/Stamp/graph is expanded, the admin clicks on the Export (PDF and Excel) option available in the expanded view. a) If the admin selects PDF, the system generates a snapshot of the current screen, including the expanded widget/Stamp/graph, and prepares the file for download. The system ensures that the entire expanded view is captured as a static image in the PDF file. b) If the admin selects Excel, the system exports the data displayed in the expanded widget/Stamp/graph in a grid format (tabular format). The system ensures that all relevant data points are included and accurately represented in the Excel file, allowing for further analysis or manipulation. 10) No Data Available: If no data exists for the selected date range, the system displays a message: "No data available for the selected period".
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	11) Export Failure: If the export function fails, the system displays an error message and prompts the admin to retry.
Field Validations	NA.
Business Validations	1) The last 7 days would be the default filter for date range. 2) Sorting and expanding widgets must not affect data integrity, ensuring consistency across all displayed metrics. 3) Clicking on Stamps or widgets should display accurate data and related details, including call transcriptions.
Notes	1) Ensure that Stamps and widgets load efficiently, regardless of the data volume. 2) Provide tooltips for metrics to clarify their significance and usage. 3) Consider implementing caching for frequently accessed data to improve Dashboard performance.

5.3 Customers

This module allows the admin to view, add, and manage customer records, as well as send subscription payment links to new customers.	
Preconditions	1) The system must have an active Stripe link for payment processing. 2) A minimum of one subscription plan needs to be available in the Subscription Plans module with Active status. 3) The list of all countries should be stored in the backend.
Post Conditions	1) New customer records are successfully added and displayed in the customer listing grid. 2) Admins can view and manage customer details, edit records, and send payment links. 3) Customers receive registration and payment confirmation emails, along with account setup instructions.
Normal Flows	1) The admin logs into the Admin Portal and selects the Customers module. 2) The system displays a customer listing screen with the following columns in a grid: a) Customer (Name of the enterprise or the main outlet) b) Address: Combination of Town/City + Country (London – UK). c) Plan d) Campaigns e) Users f) End Date g) Status h) Beta (Toggle: Yes or No) 3) To add a new customer, the admin clicks the + Add New button. 4) In the entry form, the admin needs to enter the following information and click the Publish button: a) Customer Name b) AO Name c) AO Phone Number d) AO Email Address e) Address

	f) Subscription Plan (After selecting the subscription plan from the dropdown, the admin needs to enter the price for the selected plan) 5) A confirmation popup is displayed with the message: "Have you shared the contract with the customer?". The popup provides two options: Yes and No. 6) If the admin selects Yes, the system confirms the successful addition of the new customer with a success message and provides an option to send a Stripe payment link. If the admin selects No, the system will save the information as a draft without showing the stripe payment link. 7) The admin clicks the Stripe Payment Link option. 8) The system sends a payment email with a Stripe payment link to the customer. 9) When the customer clicks the link, they are redirected to the payment page to complete the transaction. 10) Upon successful payment, Stripe sends a confirmation email to the customer. 11) Simultaneously, the system sends a welcome email with password setup instructions to the account owner. 12) While adding a new customer, the admin can group that customer as needed. 13) The admin can activate the beta functionality for a customer by toggling the option within the screen. 14) Upon enabling, the system will prompt a confirmation message: "Do you really want to enable the Beta functionality for this customer?" with two options: Yes and No. 15) If the admin selects Yes, the system will trigger an in-app notification and send a system-generated email to the registered email address of the account owner, outlining the new features included in the beta functionality. 16) The customer can click on the Save button to draft the information while adding a new customer. 17) The admin can mark the customer as a favourite (up to 5). 18) Customer Fails to Complete Payment: If the customer does not complete payment, the system should prompt the admin to resend the payment link or follow up manually. 19) While editing (by clicking a record) an existing record, the following tabs will be visible: a) Outlets: This would be the listing screen of all the outlets associated with the respective customer with the following columns bound in a grid: • Outlet • Phone Number • Canari Number • Users • Campaigns
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	• Groups • Status • The first section of this screen will showcase the following widgets: ○ Total Phone Numbers (Active + Available) ○ Remaining Minutes • The admin can add an outlet as per the customer's request from this screen and the same outlet will be reflected in the Outlets screen of the Customer portal. When adding a new outlet, the admin can enter the onboarding questionnaire based on the customer's discussion. • The admin can see the dashboard information for the selected outlet by clicking on any records. b) Users: This would be the listing screen of all the users associated with the customer. The following columns would be bound in a grid: • Name • Email Address • Last Login • Status • The admin can add a new user by clicking the "+ Add New" button. • The admin can filter or search the information by using the dedicated buttons. • While adding a new user, the admin can tag them into a group or outlet. c) Call Transcriptions: In this tab, the admin can view all the call transcriptions associated with the selected customer in a listing screen. The following columns would be bound in a grid: • Call ID • Caller Number • Date & Time • Number (To) • Duration • Type (Outlet/Campaigns) • By clicking on any record, the admin will be able to view more analytics about that call along with an option to see the whole transcription. • The admin can filter or search the information from the dedicated buttons. d) Dashboard: In this tab, the dashboard's analytics information will be showcased as per the selected customer. The admin can use the search box to search for a specific group or outlet. The system will display all relevant search results. Upon selecting a group or outlet, this screen will update to display information based on the selected group or outlet.
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	e) Campaigns: In this tab, all the campaigns from all the outlets will be reflected in the listing screen with the following columns bound in a grid: • Title • Canari Number • Start Date • Total Bookings (6 bookings/100 calls) • Tags • ROI • Status • The admin can filter or search the information from the dedicated buttons. • The admin can add a new campaign as per the customer's request from this screen (by clicking the + Add New button) and the same information will be reflected in the Campaigns module (customer portal). f) Events: In this tab, all the events associated with the selected customer will be reflected in the listing screen with the following columns bound in a grid: • Title • Start Date • End Date • Outlets (up to 2, with remaining displayed as "+ X more") • Tags (up to 2, with remaining displayed as "+ X more") • Status • The admin can filter or search a specific information from the dedicated buttons. • The admin can add a new event as per the customer's request from this screen (by clicking the + Add New button) and the same information will be reflected in the Events module (customer portal). g) Canari Numbers: This would be the listing screen of all the phone numbers associated with the selected customer. In the listing screen, the following columns would be reflected bound in a grid: • Canari Number • Outlet • Campaigns • Type (Subscription or Additional) • Used Minutes • Status • Above the listing screen, there would be the following widgets: ○ Total Numbers (Allocated + Available) ○ Average Call Time (This month) ○ Average Call Time/Campaign (This month)
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	- The admin can select the date timeframe and the screen will refresh to showcase the updated information as per the selected timeframe. h) Ask Canari: In this tab, there will be a summary (text format) of all questions asked in the customer portal for the selected customer along with all those questions and respective answers (This screen will have multiple tabs associated with an outlet). i) Billing: This would be the listing screen of all the invoices associated with the respective customer directly fetched from Stripe with the following columns bound in a grid: - Invoice Number - Invoice Date - Invoice Amount - Type (e.g., Subscription Plan, New Phone Number, Additional Minutes etc.) - Paid Date - Download Option
Field Validations	Adding a new customer: - Customer Name: Textbox, alphanumeric, special characters supported, maximum length = 250 characters and mandatory. For single outlet customers, the customer name would be equal to the outlet name. - AO Name: Textbox, alphanumeric, special characters supported, maximum length = 250 characters and mandatory. - AO Phone Number: Textbox with ISD dropdown, numeric, maximum length = 15 characters and mandatory. - AO Email Address: Textbox, alphanumeric, format specific (such as <u>user@domain.com</u>), mandatory and maximum length = 250 characters. This would be the email address to log into the customer portal for the account owner. - Country: Drop-down list and mandatory. The options would be populated as per the country list available in the backend. - Address Line 1: Textbox, alphanumeric, special characters supported, maximum length = 250 characters and mandatory. - Address Line 2: Textbox, alphanumeric, special characters supported, maximum length = 250 characters and mandatory. - City: Textbox, alphanumeric, special characters supported, maximum length = 250 characters and mandatory. - State/Province/Region: Textbox, alphanumeric, special characters supported, maximum length = 250 characters and mandatory. - ZIP/Postal Code: Textbox, alphanumeric, maximum length = 10 characters and mandatory. - Subscription Plan: A dropdown list and the options would be as per the Subscription Plans (from the Subscriptions module) and mandatory. Once selected, an inline editable grid with the following columns will appear:

	a) Phone Number(s): Read-only field as per selected subscription plan b) Outlet(s): Read-only field as per selected subscription plan c) User(s): Read-only field as per selected subscription plan d) Minutes: Read-only field as per selected subscription plan e) Plan Value: Numeric (without decimal and should be > 0), in USD (\$) and mandatory 12) Group(s): Text area with comma-separated functionality, maximum length = 1000 characters and optional. 13) Status: Two options: Active or Inactive. The system will automatically calculate as per the End Date and it will be a read-only field. 14) Description: Text area, alphanumeric along with special character supported, maximum length = 1000 characters and optional. Adding a new outlet: 1) Outlet Name: Textbox, alphanumeric, special characters supported, maximum length = 250 characters and mandatory. 2) Outlet Phone Number: Textbox with ISD dropdown, numeric, maximum length = 15 characters and mandatory. 3) Canari Phone Number: Dropdown list and mandatory. The options would be as per the system. 4) Country: Drop-down list and mandatory. The options would be populated as per the country list available in the backend. 5) Address Line 1: Textbox, alphanumeric, special characters supported, maximum length = 250 characters and mandatory. 6) Address Line 2: Textbox, alphanumeric, special characters supported, maximum length = 250 characters and mandatory. 7) City: Textbox, alphanumeric, special characters supported, maximum length = 250 characters and mandatory. 8) State/Province/Region: Textbox, alphanumeric, special characters supported, maximum length = 250 characters and mandatory. 9) ZIP/Postal Code: Textbox, alphanumeric, maximum length = 10 characters and mandatory. 10) Average Spend: Textbox, numeric, should be > \$0 and mandatory. 11) Visit Frequency: Textbox, alphanumeric, maximum length = 50 characters and mandatory. 12) Group Booking Size: Textbox, numeric, should be > \$0 and mandatory. 13) Type of Outlet: Textbox, alphanumeric, maximum length = 50 characters and optional. 14) Group(s): Text area with comma-separated functionality, maximum length = 1000 characters and optional. 15) Status: Mandatory and drop-down field with the following options: a) Active: The system stores all the calls and processes the analytics.
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	b) Inactive: The system has recorded all the calls and analytics before Inactive status. All the phone numbers associated with this outlet are available to use. The customer can activate this outlet whenever needed and the system will start calculating the analytics. c) Expired: The system has recorded all the calls and analytics before the Expired status. All the phone numbers associated with this outlet are now available to use and the customer won't be able to activate this outlet for analytics purposes. 16) Opening Hours: The outlet's opening time for each day will be captured by selecting checkboxes and drop-downs along with an option to add break time (if applicable). 17) Description: Text area, alphanumeric along with special character supported, maximum length = 1000 characters and optional. Adding a new user: 1) Name: Textbox, alphanumeric, special characters supported, maximum length = 250 characters and mandatory. 2) Phone Number: Textbox with ISD dropdown, numeric, maximum length = 15 characters and mandatory. 3) Email Address: Textbox, alphanumeric, format specific (such as <u>user@domain.com</u>), mandatory and maximum length = 250 characters. 4) Outlet(s): Multi-select dropdown (options would be as per Outlets) and optional. 5) Permission(s): A mandatory dedicated section with all the modules/sub-modules/sections of the customer portal with the following columns (the options would be shown as a toggle with Yes and No): a) View: This will enable the respective user to view only access. The user won't be able to add any details or modify any existing records. b) Edit: This will enable the respective user to edit access. The user will be able to add new details along with modification in existing records. 6) Status: Toggle with options: Active and Inactive. This would be a mandatory field.
Business Validations	1) All new customers must complete registration and make the subscription payment to access the customer portal. 2) An automated follow-up email will be triggered for customers who haven't completed payment within 3 days. 3) While adding a new customer, the default value for the usage limit will be set to 90% when the auto top-up feature is activated. The auto top-up minutes added to the customer's account will be calculated as 10% of the total minutes in the selected subscription plan. The admin needs to set up the amount to be charged for the top-up minutes, by default it would be \$0.1/minute. The admin has

	the flexibility to modify these settings at any time. All the fields would be mandatory and should be > 0. 4) All additional or FOC minutes will expire according to the end date of the customer's subscription plan. 5) The system will adhere to the following priority order when allocating minutes to a customer: a) Subscription Plan Minutes: The primary priority will be the minutes allocated as part of the customer's active subscription plan. b) Purchased Additional Minutes: The second priority will be given to any additional minutes purchased by the customer beyond the standard subscription allocation. c) Admin-Granted Free Minutes: The third priority will be assigned to any additional minutes provided to the customer by the admin at no additional cost. d) Auto Top-Up Minutes: The final priority will be for any auto top-up minutes, which are allocated when the customer's allocated minutes are exhausted. 6) The Canari Phone Number field must be unique to each customer to prevent duplicate records. 7) On the customer listing screen, the system shall display the total number of active customers by default, enclosed in parentheses. Similarly, on the outlets, users, and campaigns listing screens, the system shall display the total number of active outlets/users/campaigns by default, enclosed in parentheses. 8) The admin can assign free of cost additional minutes to any customer. The respective customer will get an in-app notification and a system-generated email regarding the same. 9) If a customer record is inactive, restrict access to the customer portal until payment is confirmed.
Note	1) Ensure the customer listing grid is searchable and filterable by attributes such as Status, Subscription Plan, Canari Phone Number etc.

5.4 Subscriptions

This module allows administrators to create, edit, and manage subscription information, and notify customers of any changes to active plans.	
Precondition	1) The admin must be logged into the Admin Portal with permission to create and manage subscription plans.
Post Conditions	1) New plans are successfully added to the system and displayed in the listing grid. 2) Customers associated with an active plan receive automatic email notifications if changes are made to the subscription details.
Normal Flows	1) The admin logs into the Admin Portal and selects the Subscriptions module. 2) The system will showcase a listing screen with 3 tabs.

	3) First Tab (Subscription Plans): A listing screen with a grid showing the following columns: a) Plan Name b) Features c) Start Date d) Active Customers e) Status 4) Second Tab (Additional Number Plans): A listing screen with a grid showing the following columns: a) Features b) Start Date c) Active Customers d) Status 5) Third Tab (Additional Minutes Plans): A listing screen with a grid showing the following columns: a) Features b) Start Date c) Active Customers d) Status 6) To add a new plan, the admin clicks the + Add New button (in the respective tab). 7) An entry form appears, allowing the admin to input details for the new plan. 8) After filling in the details, the admin clicks the Publish button. 9) The system confirms that the respective plan has been published and displays it in the listing grid. 10) The admin can filter the listing screen information as per the date range and export the data in two formats: Excel and PDF. 11) The admin can click the Save button to store the information in the system without making it available to the customers. 12) Edit Existing Plan: a) The admin clicks on a plan from the grid. b) The admin makes the necessary changes and clicks Publish. c) The system auto-generates an email to notify all customers associated with the updated plan details.
Field Validations	1) Plan Name: Textbox, alphanumeric along with special characters supported, maximum length = 250 characters and mandatory. 2) Features (Subscription Plan): An inline editable grid with the following columns: a) Phone Number(s): Numeric (without decimal and should be > 0) and mandatory b) Outlet(s): Numeric (without decimal and should be > 0) and mandatory c) User(s): Numeric (without decimal and should be > 0) and mandatory d) Minutes: Numeric (without decimal and should be > 0) and mandatory

	e) Plan Value: This would be blank as the admin will enter this information as per the customer 3) Features (Additional Numbers Plans): An inline editable grid with the following columns: a) Phone Numbers: Numeric (without decimal and should be > 0) and mandatory b) Plan Value: Numeric (without decimal and should be > 0), and mandatory 4) Features (Additional Minutes Plan): An inline editable grid with the following columns: a) Minutes: Numeric (without decimal and should be > 0) and mandatory b) Plan Value: Numeric (without decimal and should be > 0), and mandatory 5) Start Date: Date picker and mandatory. 6) End Date: Date picker and optional. This would be mandatory only to change the status to Inactive. 7) Status: Read-only textbox with the following options: a) Draft: When the admin clicks the Save button. b) Pending: When the start date > the current date. c) Active: When the start date ≤ the current date and the end date is either blank or a future date. d) Expired: When the expired date is a past date. 8) Active Customers: Read-only textbox and formula = Sum of active customers associated with the selected plan. 9) Description: Text area, alphanumeric along with special character supported, maximum length = 1000 characters and optional.
Business Validations	1) All the plan value will be in USD (\$) only. 2) There would be two types of subscription plans: Monthly and Annual. The total allocated minutes for each customer will be assigned upfront, based on the selected subscription plan, with the validity period extending until the end of the respective month or year, depending on the plan type. 3) Any additional phone numbers added to the customer's account will incur a recurring monthly/annual charge as per the existing subscription plan. 4) Customers may purchase additional minutes as an add-on to their subscription. The charge for these additional minutes will be applied once at the time of purchase. These additional minutes will remain valid until the end of the subscription plan's term (either monthly or annual), in line with the original plan's expiration date. 5) Any changes to active plans trigger an automated email notification to customers associated with that plan to ensure transparency. 6) The end date must be later than the start date for any subscription plan. 7) The admin cannot edit an Expired plan.
Notes	1) Ensure that all plan updates are logged for tracking and auditing purposes.

	2) The admin should periodically review active plans to ensure all information remains accurate and relevant.
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5.5 Ask Canari

This module enables the admin to query system data and review customer portal searches, providing insights into customer interactions and supporting admin queries through a Chat GPT-like interface.	
Preconditions	1) The admin must be logged into the Admin Portal with permission to access the Ask Canari module. 2) The system must have data on customer searches and responses in the Customer Portal. 3) Permissions must be configured to determine access to the customer details tab. 4) Customer feedback on satisfaction scores must be collected and stored.
Post Conditions	1) The admin can query system data and obtain responses for module-specific analytics. 2) The admin can view all customer queries and associated responses in a card-based interface. 3) Detailed search history for individual outlets is displayed in separate tabs. 4) The admin can filter queries by date range or search specific questions (For the Customers).
Normal Flows	1) The admin logs into the Admin Portal and navigates to Ask Canari. 2) The system opens with two tabs: a) First Tab (Default): • The system will provide a set of pre-configured prompts that the admin can select from. Upon selecting any of these prompts, the system will generate results based on the chosen prompt, ensuring a tailored response according to the selected query. • Chat GPT-like interface is displayed, allowing the admin to: ○ Enter or speak queries related to analytics or portal functionalities. ○ View responses in a conversational format. • The admin can type a query to ask a question. • The system processes the query and responds to text format. • The admin uses the Read Aloud feature to listen to the response if preferred. b) Second Tab (Permission specific): • Displays customer details in a card view. • Each card includes: ○ Customer Name ○ Total Outlets ○ Total Users ○ Top 5 Searches (filtered by selected date range)

	○ Satisfaction Score (based on customer feedback). 3) The admin clicks on a customer card to view detailed searches for all outlets in separate tabs: a) Each tab displays: • A summary section with the top 5 searches for the outlet. • A Chat GPT-like interface showing individual searches and responses. b) The admin can select a date range or search for specific queries using the tools available at the top-right corner. 4) Unsupported Query: If the admin asks a question outside the system's scope (e.g., general knowledge), the system responds with: "I can only answer questions related to your outlet's analytics". 5) No Data Available: If no queries are available for a customer, the system displays a message: "No search data available for the selected date range". 6) Search Not Found: If a query is not found within the selected date range or keyword search, the system prompts: "No results match your search criteria". 7) Regenerating Results: The system will provide the admin with the option to regenerate a result by selecting the dedicated Regenerate button. 8) Providing Feedback on Results: When the admin marks any result with a Thumbs Down, the system will prompt the admin to provide additional notes to capture the reason for the feedback. 9) The admin can view the submitted feature requests by clicking the dedicated button in a card view format.
Field Validations	1) Query Box: Text area, alphanumeric along with special characters supported, maximum length = 1000 characters and mandatory to get system response.
Business Validations	1) The scope of this module would be text to text interaction only. 2) Upon refreshing the page, the admin will no longer have access to their previous chat history. 3) By default, cards should only display customers with active accounts in the Customer Portal. 4) Satisfaction scores must be calculated and updated in real time based on customer feedback. 5) Tabs and search histories should align with the exact outlet hierarchy associated with the selected customer.
Notes	1) Ensure the tabbed interface supports scalability for customers with multiple outlets. 2) Provide tooltips for metrics like "Satisfaction Score" to explain how it is calculated. 3) Consider implementing caching for frequently accessed queries to improve performance.

5.6 Support

This module enables the admin to view, manage, and resolve customer-raised tickets with real-time notifications and detailed tracking.	
Preconditions	1) The admin must be logged into the Admin Portal with permission to access and manage the Support module. 2) Customers must have the ability to raise tickets in the system, which are then populated in the listing screen. 3) To streamline the management of support tickets raised through the customer portal, a dedicated email address (e.g., support@canari) will be established for the Admin Portal. All support tickets generated by customers will be automatically routed to this designated inbox.
Post Conditions	1) Admins can view and manage customer tickets, updating their statuses and adding details if required. 2) Customers receive real-time in-app and email notifications whenever ticket updates occur.
Normal Flows	1) The admin logs into the Admin Portal and navigates to the Support module. The system displays a listing screen with columns: a) Ticket ID b) Customer Name c) Title d) Submitted On e) Status 2) The admin clicks a record to view additional details, including: a) Outlet Name b) Raised By (along with date and time Stamp) 3) The admin updates the ticket status by: a) Clicking the record. b) Modifying the status field. c) Adding a description or attachments as needed. d) Publishing the changes. 4) The system notifies the customer in real-time through in-app and email notifications about the update. 5) The admin can search for specific information by using the search button. 6) At the top of the screen, the admin clicks on the Feature Requests button. 7) Upon clicking that button, the system displays a list of all feature requests submitted by customers in a card view format. 8) The admin clicks on any of the request cards to view further details about the specific feature request. 9) The system displays the following information for the selected feature request: a) Request ID: A unique identification number associated with the respective feature request. b) Title: The title of the feature request.

	c) Description: A detailed description of the feature request submitted by the customer. d) Requested By: The name of the user (along with the associated customer and outlet name) who submitted the request, along with the date and time Stamp of when the request was made. 10) The admin can update the record without publishing it (to the customer) by clicking on the Save button. 11) No Tickets Found: If there are no tickets in the system, the listing screen displays a message: "No tickets available". 12) Failed Update: If the admin fails to publish ticket updates due to a system error, the system logs the error and prompts the admin to retry.
Field Validations	While updating a ticket: 1) Status: Single selection dropdown and mandatory. The following would be the options: a) Open: The support ticket has been created but has not yet been addressed. The issue is actively awaiting for initial response from the support team. b) In Progress: The support team is actively working on resolving the issue. The ticket is being investigated, and actions are being taken toward a resolution. c) Closed: The ticket has been resolved and no further action is needed. The support team has marked the ticket as closed. 2) Description: Text area, alphanumeric along with special characters supported and attachment functionalities, maximum length = 1000 characters and mandatory. The following would be the format of the attachment(s): a) Image: Format: JPEG, GIF and PNG. Maximum size: 10 MB.
Business Validations	1) Status updates must be logged with timeStamps and admin details for tracking. 2) Notifications to customers should include the ticket ID and updated status to ensure clarity.
Note	1) Ensure that attachments are scanned for viruses or malicious content before upload.

5.7 Settings

5.7.1 Account

This section allows admin users to view and manage their personal information, with updates reflected across the system and email notifications sent upon changes.	
Preconditions	1) The admin must be logged into the Admin Portal with access to the Settings module. 2) The system must store and retrieve the admin's personal information (e.g., name, email, phone number).
Post Conditions	1) The admin's updated personal information is saved in the system and reflected in the Users section.

	2) A notification email is sent to the registered email address confirming the changes.
Normal Flows	1) The admin logs into the Admin Portal and navigates to Settings > Account. 2) The system displays the admin's personal information, including Name, Email Address, Phone Number etc. 3) The admin will have the ability to configure the refresh time for the "Non-Core components" by inputting the desired refresh time (in the dedicated text box) and confirm the change by clicking the Submit button. 4) Upon clicking the Submit button, the system will begin refreshing all Non-Core components for all customers on the customer portal according to the newly set frequency, starting from the following day. 5) The admin updates any of the available fields and clicks Save. 6) The system validates the updated information and updates it in the database. 7) The changes are reflected in the Users section, and the admin receives a confirmation email. 8) Incomplete Fields: If required fields (e.g., Name, Email) are left blank, the system prompts the admin to complete the missing fields before saving. 9) Permission Restricted Users: Admin users without full permissions can only view and update their own information.
Field Validation	1) Name: Textbox, alphanumeric along with special characters supported, maximum length = 250 characters and mandatory. By default, it would reflect the information as per Users section. 2) Email Address: Textbox, alphanumeric, format specific (such as user@domain.com), maximum length = 250 characters and mandatory. By default, it would reflect the information as per the Users section 3) Phone Number: Textbox with ISD dropdown, numeric, maximum length = 15 characters and mandatory. By default, it would reflect the information as per the Users section
Business Validations	1) Only authorised users with full permissions can view or edit other users' personal information. 2) Changes made in the Account section must be synchronised with the Users section for consistency. 3) An email notification must be sent to the registered email address for any updates made.
Note	1) Ensure that sensitive personal information is encrypted both at rest and in transit.

5.7.2 Roles

This submodule allows admins to create, edit, and manage user roles, including viewing assigned permissions and version history for role changes.

Preconditions	1) The admin must be logged into the Admin Portal with sufficient permissions to create and manage roles. 2) The system should have a predefined list of permissions available for assigning roles.
Post Conditions	1) New roles are successfully added and displayed in the roles listing grid. 2) The admin can view and update role permissions and manage role assignments. 3) Any changes in permissions are recorded in the version history for auditing.
Normal Flows	1) The admin navigates to the Settings and selects the Roles. 2) The system displays a roles listing screen with the following columns: a) Name b) Description c) User Count d) Status e) Permission 3) To add a new role, the admin clicks the + Add New button. 4) An entry form appears, allowing the admin to input role details (e.g., Name, Description, Permissions etc.). 5) After completing the form, the admin clicks Publish to add the role. 6) The system displays a confirmation message, and the new role appears in the grid. 7) To view role permissions, the admin clicks the designated icon under the Permission column, displaying the permissions for that role. 8) The admin can click on the Save button to store the information without making it available to use. 9) Edit Existing Roles: a) The admin clicks an existing role from the grid to modify the role's details or permissions. b) After making changes, the admin clicks Save, and the system confirms the update and logs the changes in the version history. 10) If the admin lacks permission to create or edit roles, the system displays a message indicating restricted access. 11) The system will display a validation message if the admin wants to change the status from Active to Inactive of a role having a user count of more than 1.
Field Validations	1) Name: Textbox, alphanumeric along with special characters supported, maximum length = 250 characters and mandatory. 2) Description: Text area, alphanumeric along with special characters supported, maximum length = 1000 characters and mandatory. 3) User Count: Textbox, numeric, and in read-only mode. The system will calculate as per the Users module roles filed.

	4) Status: Toggle with two options: Active and Inactive and mandatory. By default, in the entry form, it will be showcased as Active. 5) Permission: Separate section with all the module/sub-module/section names with two columns: Add and View. The options would be shown as a toggle with Yes and No.
Business Validations	1) All roles must have unique names to avoid duplication in the system. 2) Add permission will automatically enable the View permission in the entry form. 3) Without enabling a minimum of 1 View permission for any module/sub-module, the admin won't be able to save the records. 4) Only Active and Published roles need to appear as a drop-down option in the Users submodule. 5) The Admin user role will be created from the backend and cannot be editable. 6) Only authorised admins should be able to add or modify roles to ensure secure access management.
Note	1) Ensure the roles grid is searchable and filterable to help admins quickly locate specific roles.

5.7.3 Users

This submodule enables admins to add and manage user accounts within the Admin Portal, allowing them to assign roles and configure user access permissions.	
Preconditions	1) The admin must be logged into the Admin Portal with permission to add and manage users. 2) The system should have predefined roles (in the Roles) that can be assigned to new users.
Post Conditions	1) New user accounts are successfully added to the system and displayed in the user listing grid. 2) The admin can update user details and assign roles as needed. 3) The admin can manage user status to activate or deactivate access within the Admin Portal.
Normal Flows	1) The admin navigates to the Settings module and selects the Users. 2) The system displays a user listing grid with the following columns: a) Name b) Phone Number c) Email Address d) Designation e) Roles f) Last Login g) Status 3) To add a new user, the admin clicks the + Add New button. 4) An entry form appears, allowing the admin to input the new user's details. 5) After completing the form, the admin clicks Publish to add the user to the portal.

	6) The system displays a confirmation message, and the new user appears in the user listing grid. 7) The admin can click on the Save button to store the information without making it available to use. 8) Edit User Record: a) The admin selects an existing user by clicking on the record in the grid. b) After making the necessary changes, the admin clicks Publish, and the system confirms the update. 9) If the admin lacks permission to add or edit users, the system displays a message indicating restricted access.
Field Validations	1) Name: Textbox, alphanumeric, special characters supported, maximum length = 250 characters and mandatory. 2) Phone Number: Textbox with ISD dropdown, numeric, maximum length = 15 characters and mandatory. 3) Email Address: Textbox, alphanumeric, format specific (such as <u>user@domain.com</u>), mandatory and maximum length = 250 characters. 4) Designation: Textbox, alphanumeric along with special characters supported, maximum length = 250 characters and optional. 5) Roles: Multi-select dropdown list and the options would be as per the Roles submodule and mandatory. 6) Status: Toggle with options: Active and Inactive. 7) Description: Text area, alphanumeric along with special characters supported, maximum length = 1000 characters and option.
Business Validations	1) Each user must have a unique email address within the system to prevent duplicate records. 2) Any changes to a user's role or status should be logged for tracking purposes. 3) A Sysadmin user will be added from the backend with admin role rights and cannot be editable. 4) Only the Active and Published roles need to appear as drop-down options for the Roles field.
Notes	1) Ensure the Users grid is searchable and filterable by attributes such as Role, Status, and Designation for easier navigation. 2) Implement audit logging for changes in user roles and statuses to maintain a record of all modifications.

5.7.4 Calendars

This section enables the admin to view, manage, and mark holidays or any special occasion for specific regions, linking associated calls during these periods for analytics.	
Preconditions	1) The admin must be logged into the Admin Portal with access to the Calendars section. 2) Third-party APIs must be integrated to fetch holiday/occasion data for all regions. 3) Outlets must be pre-configured in the system.

Post Conditions	1) Holidays and occasions are added to the calendar and displayed on the home screen. 2) Calls made during these periods are linked to the respective occasion for insights and analytics. 3) Occasions are applied to all outlets in the selected region(s).
Normal Flows	1) Viewing Holidays/Events: a) The admin navigates to Settings > Calendars. b) The system displays a full-screen calendar view showing holidays/occasions from all regions (default: current month). c) The admin clicks on a specific date to view details of holidays/occasions for that date. 2) Adding Holidays/Events: a) The admin clicks on the + Add New button at the top-right corner. b) The system displays a pop-up form with the following fields: • Title: Short description of the event. • Date Range: Select start and end dates (future dates only). • Region(s): Multi-select dropdown to choose applicable regions. c) The admin fills out the form and clicks Publish. d) The system confirms the action and displays the event on the calendar view. 3) Manual Refresh: a) A button for manual refresh will be located in the top-right corner. b) Upon clicking the manual refresh button, the system will prompt the admin to select one or more regions from a multi-select dropdown list. c) After the desired region(s) are selected, the admin will click the Submit button. Following this, the system will update the calendar details for all outlets within the selected region(s), ensuring that the most current data is reflected across all relevant locations. 4) The admin uses the filter to locate specific occasions or holidays by date, title, or region.
Field Validations	1) Title: Textbox, alphanumeric, special characters supported, maximum length = 50 characters and mandatory. 2) Date(s): Date picker (start and end), $\geq$ current date, and mandatory. 3) Region(s): Multi-select dropdown and mandatory. The options would be fetched from the backend.
Business Validations	1) The system will automatically refresh the calendar details for all outlets across all regions every Monday. 2) Occasions must not overlap with existing occasions for the same region(s). 3) Past dates cannot be edited or marked as holidays/occasions. 4) Calls during marked event periods must be correctly linked to the respective occasions for analytics.

	5) The system must handle occasion applicability for all associated outlets within the selected region(s).
Notes	1) Ensure that holidays/occasions fetched from third-party APIs are displayed with proper tagging to distinguish them from admin-created events. 2) Add tooltips or labels for calendar icons to improve usability. 3) Implement reminders for upcoming holidays/events to notify admins for better planning.

5.7.5 Billing

This section enables the admin to view and manage billing details for all customers, with the option to download invoices and filter data using interactive widgets.	
Preconditions	1) The admin must be logged into the Admin Portal with permission to access the Settings module. 2) The system must integrate with Stripe for billing records linked to all customers.
Post Conditions	1) The admin can view all customer billing details on a listing screen. 2) Invoices can be downloaded as a PDF. 3) Widgets dynamically filter the listing screen based on selected metrics.
Normal Flows	1) The admin logs into the Admin Portal and navigates to Settings > Billing . 2) The system displays widgets showing metrics, such as: - Total Active Customers - New Customers - Expiry in Next 3 Months - Total Invoiced Amount 3) Below the widgets, the system displays a listing grid with columns: - Customer Name - Invoice Number - Invoice Date - Type (Subscription Plan, Additional phone Number or Additional Minutes) - Invoice Amount 4) All the information of the widgets and listing screen will be showcased as per the selected date range. 5) The admin clicks a widget (e.g., "Expiry in Next 3 Months"), and the system filters the listing grid based on the selected metric. 6) The admin clicks on the widget to reset the screen to its default view. 7) Upon clicking the View button, the admin will be redirected to the Stripe page, where the system will display additional details for the selected invoice. The admin will also have the option to download the invoice from this page. 8) Widget Data Unavailable: If a widget metric (e.g., "New Customers") has no data, the system greys out the widget and displays a message: "No data available for this metric".

	9) Failed Invoice Download: If the invoice download fails due to a system error, the admin is prompted to retry.
Field Validation	NA
Business Validations	1) By default, all the invoices linked to active customers should be displayed on the listing grid. 2) The listing screen will prioritise displaying the details of favourite customers (as per the Customers module), followed by the details of the remaining customers. 3) Widgets must be synchronised with the database to display up-to-date metrics. 4) By default, the current month's data will be displayed on the listing screen.
Note	1) Include tooltips for widgets to help admins understand the metrics they represent.

5.7.6 Help and Resources

This section enables the admin to upload, manage, and publish contents, for customers in both the website and the customer portal.	
Preconditions	1) The admin must be logged into the Admin Portal with permission to manage this module. 2) Content must be configured to support dynamic updates and filtering.
Post Condition	1) New contents are published and visible on the website and customer portal.
Normal Flows	1) The admin logs into the Admin Portal and navigates to Settings > Help and Resources. 2) The system displays a card view of all existing contents, showing their titles and other information. 3) To add a new content: a) The admin clicks + Add New. b) An entry form appears, prompting the admin to fill in: • Title (name of the content) • Description (corresponding response along with attachment functionalities) • Type (Campaigns, Dashboard etc.) • Start Date (date for publication) c) The admin clicks Publish, and the system displays a confirmation message. d) The new content is published on the website and customer portal. 4) The system will highlight the most viewed content. 5) Unpublished Content: If the admin clicks Save instead of Publish, the information is stored in the system but not published. The admin can publish it later by editing the content and clicking Publish.

	6) Missing Required Fields: If any required fields are left blank, the system displays an error message and prevents the respective content from being saved or published.
Field Validations	1) Title: Textbox, alphanumeric, special characters supported, maximum length = 250 characters and mandatory. 2) Description: Text area, alphanumeric, special characters supported, attachment functionality, maximum length = 1000 characters and mandatory. The admin can paste a URL in the field. The attachment functionalities would be as follows: a) Image: Format: JPEG, GIF and PNG. Maximum size = 10 MB. b) Video: Format: Video URL, MP4 and WEBM. Maximum size = 20 MB. 3) Thumbnail: This would be a mandatory field, maximum size = 10 MB and Format: JPEG, GIF and PNG. 4) Type: Textbox, alphanumeric, special characters supported, maximum length = 250 characters and mandatory. 5) Start Date: Date picker and mandatory. By default, the system will capture the current date. 6) End Date: Date picker, > Start Date and mandatory. 7) Status: The system will auto-calculate as per the following criteria: a) Draft: The admin clicks the Save button while adding new content or the admin clicks the Publish button, but the start date is a future date. b) Active: The admin clicks the Publish button while adding new content and Start Date ≤ Current Date. c) Expired: The End Date < Current Date.
Business Validations	1) While adding/updating a record, the system will store the following details and showcase the same in the entry form: a) Created By: Name of the creator b) Created On: Creation date and time Stamp (20 Nov 2024, 09:00 AM) c) Last Modified By: Name of the last modifier d) Last Modified On: Last modified date and time Stamp (25 Nov 2024, 11:35 AM) 2) The admin must preview a video before publishing it to ensure content accuracy. 3) Once published, the respective content should only become visible from the specified Start Date.
Notes	1) Ensure the system supports embedding video links from third-party platforms (e.g., YouTube, Vimeo). 2) Include a confirmation dialogue before publishing to avoid unintentional updates.

6. Functional Requirement – Customer

6.1 Login

This screen enables registered customers to securely access the customer portal by entering their registered email ID and password or by using the Magic Link .	
Preconditions	1) The customer must be registered and have set up their account, including password creation. 2) The customer must listed on the Customers screen (in the Admin Portal) with Active status.
Post Conditions	1) Customers are successfully authenticated and directed to the Dashboard as the landing page. 2) The customer can initiate a password reset or Magic Link login if they are unable to remember their password.
Normal Flows	1) Upon launching the application, the customer will be prompted to access the login screen. 2) The customer enters their registered Email ID and Password into the respective fields. 3) The customer clicks the Sign In button. 4) If the credentials match an entry in the system database, the customer is authenticated and directed to the Dashboard. 5) The customer's assigned user rights determine access to specific modules/features. 6) Forgot Password Flow: a) If the customer cannot remember their password, they select the Forgot Password button. b) The customer enters their registered email address and clicks the Submit button. c) The system sends an automated email containing a password reset link. d) The customer clicks the link in the email and is directed to a screen to set and confirm a new password. e) The customer receives a confirmation email upon successful password reset. 7) Magic Link Flow: a) The customer selects Get a Magic Link on the login screen. b) The system sends an email with a one-time link to the customer's registered email. c) Upon clicking the link, the customer is redirected to the Dashboard without needing to enter the password.
Field Validations	3) Email ID: Textbox, alphanumeric, format specific (such as <u>user@domain.com</u>), mandatory and maximum length = 250 characters. 4) Password: Textbox, alphanumeric, format specific (must include minimum 1 letter, number and special characters), mandatory, minimum length = 8 characters and maximum length = 24 characters.

Business Validations	1) The customer must have completed registration, including password setup, to access the portal. 2) If the email/password combination does not exist in the database, display an error message: The provided credentials are not registered with the application. 3) Customers without a completed profile after payment cannot access the portal until the profile setup is complete. 4) Automated reminders for profile completion should be sent if the customer has not fulfilled the profile after 3 days.
Note	1) For unsuccessful login attempts, consider setting a limit on retries to prevent brute force attacks, with temporary locks after multiple failed attempts.

6.2 Dashboard

This module serves as the default screen upon login, providing customers with a comprehensive overview of call analytics, campaign performance, and key insights.	
Preconditions	1) The customer must be logged into the Customer Portal with access rights to view the dashboard. 2) The system must have data available to populate call analytics, sentiment analysis, and campaign performance metrics. 3) The customer's access level should be defined to determine accessible views.
Post Conditions	1) The customer can view key performance indicators (Stamps) based on the assigned permission. 2) Analytics are displayed across three sections, covering primary call analytics, sentiment analysis, and campaign performance. 3) Stamps and widgets are expandable, sortable, and filterable (by date).
Normal Flows	1) The customer logs into the Customer Portal, and the system redirects them to the Dashboard screen. 2) The Dashboard presents the information in the following tabs: a) First Tab (All): The information of all the outlets as per the assigned permission will be showcased in the following sections with Stamps and widgets: • First Section: An AI summary (text format) of all analytics available on the home screen. • Second Section (Widgets): Displays primary call analytics (e.g., number of upcoming bookings along with values, average response time, and missed calls along with potential revenue loss). • Third Section: Lost Opportunities, call reason, complaints section, peak call hours (along with call volumes), satisfaction rate, local vs tourist, new vs returning, and gender. By clicking the Future Prediction button, based on the selected date range, the system processes the historical data and predicts the future peak call hours.

	○ If the customer selects the Last 7 Days as the date range, the system will forecast the peak call hours for the next 7 days. ○ If the customer selects a different date range, the system will display future data corresponding to the selected range (e.g., the next 30 days, or another user-defined period). • Fourth Section (My Campaigns): High-level performance overview of the customer's favourite campaigns (up to 4), depicted in a line chart format. If the customer has fewer than four favourite campaigns or none at all, the system will automatically display the most engaged campaigns, up to a maximum of four, based on the selected date range. The graph will feature the following elements: X-axis: Upcoming bookings and Y-axis: Potential value. • Fifth Section (My Events): High-level performance overview of events (up to 4). If the customer has less than four favourite events or none at all, the system will automatically showcase the most engaged events (up to 4) based on the selected date range. b) Second tab (Campaigns): The customer will get all the analytics of all campaigns associated with all the outlets (as per the assigned permission). 3) The customer can use the search box to search for a specific outlet, group or campaign. The system will display all relevant search results. Upon selecting an option, the dashboard will update to display information based on the selected group or outlet. 4) The customer can sort and expand Stamps and widgets for detailed analysis by selecting the dedicated buttons. 5) The customer applies date filters to adjust the time range of data displayed on the dashboard. 6) The system will showcase detailed information once the customer clicks on any Stamps or widgets. 7) The customer can export the dashboard's snapshot in a print-friendly PDF. 8) While the widget/STAMP/graph is expanded, the customer clicks on the Export (PDF and Excel) option available in the expanded view. a) If the customer selects PDF, the system generates a snapshot of the current screen, including the expanded widget/STAMP/graph, and prepares the file for download. The system ensures that the entire expanded view is captured as a static image in the PDF file. b) If the customer selects Excel, the system exports the data displayed in the expanded widget/STAMP/graph in a grid format (tabular format). The system ensures that all relevant data points are included and accurately represented in the Excel file, allowing for further analysis or manipulation.
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	9) Upon the customer's first login to the customer portal, the system will present the following prompt: "Please link the Canari phone numbers to the following areas?" a) Outlet(s) b) Website c) Social Media d) Campaign 10) For each option, the system will provide a clickable link that redirects the customer to the relevant Help & Resources section, offering detailed guidance on how to complete the linking process for each area. 11) If there is insufficient data to display in any section (e.g., no recent calls or campaigns), the system displays a message stating, "No data available for the selected period". 12) Access Limitations: If a customer has restricted access to specific viewpoints (e.g., Outlet only), the system hides the unavailable viewpoints from the dashboard.
Field Validation	NA.
Business Validations	1) When a user of an outlet clicks the "Future Prediction" button, the system will prevent (if already clicked) the button from being clicked again until the next scheduled refresh cycle. This restriction ensures that the prediction is based on the most current data available. 2) During the period when the "Future Prediction" button is disabled, a View button will be displayed. By clicking on this, users will be able to access the last prediction generated, along with the timeStamp of the last refresh, to provide context on when the prediction was made. 3) During that period, when the user hovers over the "Future Prediction" button, the system will display a message informing the user that the button can be used again only after the next refresh cycle. This ensures clarity regarding the usage limitations and the timing of the next available prediction. 4) Sorting and expanding widgets must not affect data integrity, ensuring consistency across all displayed metrics. 5) All users within a customer account will share the same Stamps and widgets. However, the system will display information based on each user's assigned permission levels.
Notes	1) Ensure that each STAMP and widget is optimised for quick loading, given the potential volume of data. 2) Provide detailed tooltips on the dashboard widgets, helping customers understand each metric's significance and how it relates to their operations. 3) Implement periodic data caching for frequent metrics to enhance performance and reduce loading times for recurring dashboard views.

6.3 Campaigns

This module enables customers to create, view, and manage campaigns, allowing the system to analyse inbound calls and provide accurate campaign insights.	
Preconditions	1) The customer must be logged into the Customer Portal with permission to manage campaigns. 2) The system should be capable of storing and retrieving campaign details and linking inbound calls to active campaigns.
Post Conditions	1) New campaigns are successfully created and stored, appearing in the campaign listing grid. 2) The system begins analysing inbound calls associated with the campaign, providing insights to the customer.
Normal Flows	1) The customer logs into the Customer Portal and selects the Campaigns module. 2) A listing screen with a grid showing the following columns: a) Title b) Canari Number c) Start Date d) Total Bookings (e.g., 6 bookings/100 calls) e) Tags f) ROI g) Status 3) The customer clicks the + Add New to add a new campaign. 4) In the entry form, the following details need to be entered: a) Type (Campaign or Event) b) Select Outlet (only applicable if Type = Campaign) c) Outlet Number d) Title e) Description f) Canari Number g) Cost of Campaign h) Start Date i) Tags 5) The customer completes the form and clicks Publish. 6) The system stores the campaign details, displays a confirmation message, and updates the campaign listing grid. 7) The system begins tracking inbound calls linked to the campaign phone number, analysing call data based on campaign-specific details. 8) The customer can mark any campaign as Favourite (up to 4) by ping it. 9) Additional Phone Number/Minutes: a) The customer can initiate the process of purchasing an additional phone number/minutes by clicking the dedicated button (Buy Additional Number(s)/Buy Additional Minutes). b) Upon clicking the respective buttons, the system will retrieve the available plans from the admin portal, and display the following options on the screen:

	• Phone Number/Minutes Quantity • Amount (by default, in \$) • Validity c) Once the customer selects a plan and clicks the Buy Now button, the system will automatically process the payment in the backend according to the selected plan. d) A confirmation message will be displayed on the screen upon successful completion of the purchase. 10) The customer will have the option to store the data without publishing the campaign by clicking on the Save button. 11) Edit Existing Campaign: a) The customer clicks on an existing campaign from the grid. b) The system will showcase the campaign information in the following sections: • First Section: The following widgets will be showcased: ○ Cost of Campaign ○ Total Bookings ○ Potential Revenue ○ ROI: $(\text{Potential Revenue} - \text{Cost of Campaign}) / \text{Cost of Campaign} * 100$ • Second Section: All the entry form's details (apart from Cost of Campaign) along with the following additional information: ○ View Dashboard: Upon clicking, the system will navigate the customer to the Campaign tab on the dashboard, where the data will be filtered to display information relevant to the selected campaign exclusively. ○ Last Call Received: The system will indicate the most recent call received for the selected campaign, displaying the date and time of the last interaction. ○ Last Updated By: Name along with date and time Stamp. ○ Created By: Name along with date and time Stamp. ○ Edit: By selecting this button, the customer will be able to modify the information within this section, enabling updates to the campaign details as required. • Third Section: An AI summary (text format) of the campaign performance. c) After making changes, the customer clicks Publish to update the campaign, and the system confirms the update. 12) Upon the customer's first navigation to this module, the system will automatically display a guide video that explains the features and functionalities of the module. The customer will have the option to close the video at any time if they choose to do so. 13) If required fields are left blank, the system prompts the customer to complete the fields before saving.
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Field Validations	1) Select Outlet: Dropdown field (single selection) and mandatory. The system will showcase all the Active outlets (as per the Outlets section) as options. 2) Outlet Number: By default, it will showcase the phone number as per the Outlets section. However, the customer can edit it. This would be a mandatory field. 3) Title: Textbox, alphanumeric along with special characters supported, maximum length = 25 characters and mandatory. 4) Description: Text area, alphanumeric along with special character supported, maximum length = 1000 characters and mandatory. 5) Canari Number: Dropdown (all the available numbers as per the subscription plan and additional number), and mandatory. The last option would be showcased as "Buy Additional Number(s)". 6) Cost of Campaign: Dropdown + Textbox, numeric (without decimal and should be > 0), and mandatory. The options would be: a) One Time: The customer won't be able to edit the amount in future once published. b) Monthly: The customer will able to edit the amount in future once published. 7) Start Date: Date picker, and mandatory. 8) Tags: Mandatory text area with comma-separated functionality. By default (on the listing screen), the system will display a maximum of two tags for each campaign. If a campaign has more than two tags (e.g., six tags), the system will display the first two tags followed by a label indicating the number of additional tags (e.g., 2+4 more). The customer can hover over this label to view all the tags associated with the campaign. The system will showcase a few predefined tags by default (TBC). 9) End Date: Date picker and optional (mandatory only if need to mark the status as Expired). 10) Status: Drop-down with the following options: a) Draft: The customer did not publish the campaign. The system does not initiate analytics for campaigns with this status. b) Active: The customer published the campaign and the Start Date $\leq$ the current date. The system performs analytics calculations for campaigns with this status. c) Inactive: The system temporarily holds analytics calculations for campaigns with this status. However, upon reactivation, the system will resume analytics processing for the campaign. d) Expired: The End Date < current date. The system will no longer perform analytics for campaigns with this status. End Date is mandatory to mark the status as Expired. The customer won't be able to use this campaign in future.
Business Validations	1) Payment Method Validation (while buying new number/minutes): a) If no payment method is associated with the customer's account in Stripe, the system will display a validation message prompting the customer to add a payment method.

	b) An option will be provided to navigate to Stripe for this purpose. c) Upon clicking the provided option, the customer will be redirected to Stripe, where they can securely enter their payment details and complete the transaction. 2) By default, only the account owner will have the authority to initiate the "Buy Additional Numbers/Minutes" action. 3) If any user other than the account owner attempts to access this functionality, the system will display a validation message: "Please contact the account owner". 4) Each campaign must be associated with a unique campaign phone number to differentiate call analytics. 5) The Start Date must be set before the End Date to ensure correct tracking and analysis periods. 6) Status changes (e.g., setting a campaign to "Inactive") should stop the analysis of inbound calls for that campaign.
Notes	1) Ensure data encryption for all stored campaign information, especially phone numbers and campaign identifiers, to maintain data security. 2) Campaign data should be retained for historical reference and performance analysis, even after the campaign has ended. 3) The system should allow filtering and sorting within the grid view to help customers locate specific campaigns quickly.

6.4 Events

This module allows customers to create and manage group-wide promotions, simplifying interaction tracking and event management.	
Preconditions	1) The customer must be logged into the Customer Portal with access to the Events module. 2) Outlets and groups must be pre-configured in the system for selection during event creation.
Post Conditions	1) Events are created, saved, or published and visible on the listing screen. 2) Customers can view, edit, or mark events as favourites.
Normal Flows	1) The customer navigates to the Events module. 2) The system displays a listing screen with columns: a) Title b) Start Date c) End Date d) Outlets (up to 2, with remaining displayed as "+ X more") e) Tags (up to 2, with remaining displayed as "+ X more") f) Status 3) Creating an Event: a) The customer clicks on + Add New. b) The system displays an entry form with the following fields: • Title • Description

	• Start Date • Tags • Select Groups/Outlets • End Date c) The customer fills out the form and clicks: • Publish to save and publish the event immediately. • Save to store the data without publishing. d) The system confirms the action with a success message and displays the event on the listing screen. 4) Managing Events: a) The customer marks events as favourites (up to 4 events). b) Favourite events are displayed in the first 4 rows of the listing screen. c) The customer views or edits event details by clicking on a record. d) The details page includes: • All listing screen details • Created By with date and timeStamp • Last Updated By with date and timeStamp 5) A search bar and filters on the listing screen will be there to help customers locate specific events. 6) If no groups or outlets are available, the system displays: "No outlets/groups available for selection". 7) If the customer tries to submit the form without completing the required fields, the system prevents submission and displays an error: "Please fill in all required fields".
Field Validations	1) Title: Textbox, alphanumeric along with special characters supported, maximum length = 25 characters and mandatory. 2) Description: Text area, alphanumeric along with special character supported, maximum length = 1000 characters and mandatory. 3) Start Date: Date picker, and mandatory. 4) End Date: Date picker and optional (mandatory only if need to mark the status as Expired). 5) Tags: Mandatory text area with comma-separated functionality. By default (on the listing screen), the system will display a maximum of two tags for each event. If an event has more than two tags (e.g., six tags), the system will display the first two tags followed by a label indicating the number of additional tags (e.g., 2+4 more). The customer can hover over this label to view all the tags associated with the event. The system will showcase a few predefined tags by default (TBC). 6) Select Group(s)/Outlets: Dropdown field (multi-selection) and mandatory. The system will showcase all the Active groups/outlets (as per the Groups/Outlets section) as options. 7) Status: Drop-down with the following options: a) Draft: The customer did not publish the event.

	b) Active: The customer published the event and the Start Date $\leq$ the current date. c) Expired: The End date $<$ current date. The system will no longer perform analytics for events with this status. End Date is mandatory to mark the status as Expired.
Business Validations	1) Events must have unique titles within the same date range to prevent duplication. 2) Only active outlets and groups should appear in the selection list. 3) Caller interactions during the event period must be linked correctly for tracking purposes.
Notes	NA

6.5 Ask Canari

This module enables customers to query analytics and portal-related information through a Chat-GPT-like interface.	
Preconditions	1) The customer must be logged into the Customer Portal with access to the Ask Canari module. 2) The system should have data available such as analytics, campaigns, users, billings etc. related to the customer's account. 3) Integration for read-aloud features must be configured and operational.
Post Conditions	1) The customer receives responses to queries related to their outlet's analytics. 2) Feedback on search results is stored for future reference and improvement. 3) All queries, responses, and feedback are logged and reflected in the Admin Portal's AI Search module.
Normal Flows	1) The customer logs into the Customer Portal and navigates to the Ask Canari module. 2) The system displays a Chat GPT-like interface for entering queries. 3) The system will provide a set of pre-configured prompts that the customer can select from. Upon selecting any of these prompts, the system will generate results based on the chosen prompt, ensuring a tailored response according to the selected query. 4) The customer can type a query to ask a question. 5) The system processes the query and responds to text format. 6) The customer uses the Read Aloud feature to listen to the response if preferred. 7) The customer provides feedback on the response or regenerates the answer for clarification. 8) The system stores the query, feedback, and response in the database for future reference. 9) Unsupported Query: If the customer asks a question outside the system's scope (e.g., general knowledge), the system responds with: "I can only answer questions related to your outlet's analytics".

	10) No Data Available: If the query relates to unavailable data, the system responds with: "No data available for the requested information". 11) Regenerating Results: The system will provide the customer with the option to regenerate a result by selecting the dedicated Regenerate button. 12) Providing Feedback on Results: When the customer marks any result with a Thumbs Down, the system will prompt to provide additional notes to capture the reason for the feedback.
Field Validation	1) Query Input: Text area, alphanumeric along with special character supported, maximum length = 1000 characters and mandatory.
Business Validations	1) The scope of this module would be text-to-text interaction only. 2) Upon refreshing the page, the customer will no longer have access to their previous chat history. 3) The system should limit query responses to account-specific data and not process general knowledge queries. 4) The system will evaluate the user's access level prior to displaying any results in response to a query. 5) If User A is associated with a specific outlet, the system will restrict the search results to data associated with that outlet only. 6) If User A attempts to query data from an outlet they are not associated with, the system will display a validation message: "You do not have access to data from this outlet. Please contact your account owner for further assistance". 7) Feedback must be linked to the corresponding query and stored for analysis.
Notes	1) Ensure that all queries and responses are stored securely and comply with data protection regulations. 2) Provide a tutorial or tooltips within the module to guide customers.

6.6 Help and Resources

This section allows customers to access content and raise support tickets or feature requests through an intuitive card-based interface.	
Preconditions	1) The customer must be logged into the Customer Portal with appropriate access rights. 2) The system must have pre-configured content categories and a functional search feature.
Post Conditions	1) Customers can view and interact with content, including videos and details. 2) Customers can raise support tickets or submit feature requests, which are reflected in the respective modules in the Admin Portal. 3) Customers receive in-app and email notifications for updates to their support tickets.
Normal Flows	1) Viewing Content: - The customer navigates to Profile > Help and Resources. - The system displays a card view of content, organised into tabs based on content type.

	c) The customer clicks on a card to view associated videos and details. d) The customer uses the Search button to locate specific content. 2) Raising a Support Ticket or Feature Request: a) The customer clicks the Support button at the top of the screen. b) A pop-up form appears with the following fields: e) Type: Dropdown with options Raise a Ticket or Feature Request. f) Title: A brief of the request or issue. g) Description: Detailed explanation. c) The customer fills in the form and clicks Publish. d) The system displays a confirmation message. e) The system logs the ticket in the Support module in the Admin Portal. f) The system sends an in-app notification and email to the customer when the admin updates the ticket. 3) Feature Request: The system logs the request in the Help and Resources section in the Admin Portal. 4) No Content Available: If there is no content in the selected tab, the system displays a message: "No content available for this category". 5) Incomplete Form Submission: If the required fields in the Support form are not completed, the system displays an error: "Please fill in all required fields to proceed".
Field Validations	While raising a ticket: 1) Title: Textbox, alphanumeric along with special characters supported, maximum length = 50 characters and mandatory. 2) Description: Text area, alphanumeric along with special characters supported and attachment functionalities, maximum length = 1000 characters and mandatory. The following would be the format of the attachment(s): a) Image: Format: JPEG, GIF and PNG. Maximum size: 10 MB. b) Video: Format: Video URL, MP4 and WEBM, Maximum size: 20 MB.
Business Validations	1) Support tickets and feature requests must be tagged with unique IDs and stored in the relevant modules in the Admin Portal. 2) Notifications to customers must include relevant details, such as ticket ID, status updates, and admin comments.
Note	NA

6.7 Settings

6.7.1 Account

This section allows customers to view and manage their personal information, including email notification preferences, with updates reflected across the system.	
Preconditions	1) The customer must be logged into the Customer Portal with access to the Account section.

	The system should have stored data for each customer's personal information.
Post Conditions	- The customer's updated personal information is saved in the system and reflected in the Users section. - The updated notification frequency takes effect immediately, and the customer receives automated emails as per the new settings. - An email notification is sent to the registered email address confirming the changes.
Normal Flows	- The customer logs into the Customer Portal and navigates to Settings > Account. - The system displays the customer's personal information, including Name, Email Address, Phone Number, Email Notification Frequency etc. - The customer updates any of the available fields and clicks Save. - The system validates the updated information and saves it in the database. - The changes are reflected in the Users section, and the customer receives a confirmation email. - Incomplete Fields: If required fields (e.g., Name, Email) are left blank, the system prompts the customer to complete the missing fields before saving. - Permission Restricted Users: Customers without full permissions can only view and edit their own information.
Field Validation	- Name: Textbox, alphanumeric along with special characters supported, maximum length = 250 characters and mandatory. By default, it would reflect the information as per the Users section. - Email Address: Textbox, alphanumeric, format specific (such as <u>user@domain.com</u>), maximum length = 250 characters and mandatory. By default, it would reflect the information as per the Users section - Phone Number: Textbox with ISD dropdown, numeric, maximum length = 15 characters and mandatory. By default, it would reflect the information as per the Users section - Email Notification Frequency: Event options dropdown (such as dashboard summary) along frequency options dropdown (such as daily, weekly etc.) and mandatory.
Business Validations	- The Cancel Subscription option will be available exclusively to the account owner for the respective customer. - Upon selecting this option, the system will display a confirmation pop-up message with the following prompt: "Are you sure you want to cancel the subscription?" - The account owner will have two options: Yes and No. - If the account owner selects Yes, the system will stop billing the customer after the subscription end date and will suspend analytics services for that customer. - All the users of that customer will continue to have access to the customer portal until the end of the current subscription term.

	6) Customers can only update their own information; permission-restricted users cannot modify other's data. 7) Changes in notification frequency should immediately adjust the email scheduling for the respective user. 8) Updated information must be synchronised with the Users section to ensure consistency. 9) If a customer has not logged in for 5 days, the system will: a) Send an email containing a dashboard snapshot for the past 5 days. Ensure the dashboard information is personalised based on the user's permissions: • If the user manages multiple outlets, only associated outlets' information will be displayed. • For the account owner, the summary email will include information for all outlets under his/her account.
Note	1) Ensure all personal data is encrypted both at rest and in transit to comply with data protection regulations.

6.7.2 Users

This section enables the customer to add and manage user accounts within the Customer Portal, allowing them to configure user access permissions.	
Precondition	1) The customer must be logged into the Customer Portal with permission to add and manage users.
Post Conditions	1) New user accounts are successfully added to the system and displayed in the user listing grid. 2) The customer can update user details and assign permissions as needed. 3) The customer can manage user status to activate or deactivate access within the Customer Portal.
Normal Flows	1) The customer logs into the customer portal and navigates to Settings > Users. 2) The system displays a user listing grid with the following columns: a) Name b) Email Address c) Last Login d) Status 3) To add a new user, the customer clicks the + Add New button. 4) An entry form appears, allowing the customer to input the new user's details. 5) After completing the form, the customer clicks Publish to add the user to the portal. 6) The system displays a confirmation message, and the new user appears in the user listing grid. 7) The system will send an email to the registered email address to set up the password. 8) Edit User Record: a) The customer selects an existing user by clicking on the record in the grid.

	b) After making the necessary changes, the customer clicks Publish, and the system confirms the update. 9) The customer can click the Save button to store the information without publishing it. 10) If the customer lacks the permissions to add or edit users, the system displays a message indicating restricted access.
Field Validations	1) Name: Textbox, alphanumeric, special characters supported, maximum length = 250 characters and mandatory. 2) Phone Number: Textbox with ISD dropdown, numeric, maximum length = 15 characters and mandatory. 3) Email Address: Textbox, alphanumeric, format specific (such as <u>user@domain.com</u>), mandatory and maximum length = 250 characters. 4) Outlets: Multi-select drop-down (options would be as per Outlets) and optional. 5) Designation: Textbox, alphanumeric along with special characters supported, maximum length = 250 characters and optional. 6) Permissions: A mandatory dedicated section with all the module/sub-module/section with the following columns (the options would be shown as a toggle with Yes and No): a) View: This will enable the customer to view only access. The customer won't be able to add any new details or modify any existing records. b) Edit: This will enable the customer the edit access. The customer will be able to add new details along with modification in any existing records. 7) Status: Toggle with options: Active and Inactive. 8) Description: Text area, alphanumeric along with special characters supported, maximum length = 1000 characters and optional.
Business Validations	1) Each user must have a unique email address within the system to prevent duplicate records. 2) The Add permission will automatically enable the View permission. 3) Without enabling a minimum of 1 View permission for any module/sub-module/section, the customer won't be able to save/publish the records.
Notes	1) Ensure the Users grid is searchable and filterable by attributes such as Role, Status, and Designation for easier navigation. 2) Implement audit logging for changes in user roles and statuses to maintain a record of all modifications.

6.7.3 Outlets

This section enables the customer to set up outlet details, and designate specific outlet as Favourite for easy access.	
Precondition	1) The customer must be logged into the Customer Portal with appropriate permissions to access and manage Outlets. 2) The system must have active outlets associated with the customer's account.

Post Conditions	1) New outlets are created and displayed in the listing grid. 2) Favourite outlets appear prominently in the listing, following the remaining outlets. 3) First-time users of the outlets receive setup instructions via email.
Normal Flows	1) The customer logs into the Customer Portal and navigates to Settings > Outlets. 2) The system displays a listing grid with columns: a) Outlet Name b) Outlet Phone Number c) Canari Phone Number d) Address (combination of Town/City + Country, London – UK) e) Status 3) To add a new profile, the customer clicks + Add New button. 4) An entry form appears, where the customer fills in the outlet details and clicks Publish. 5) Upon publishing, the outlet appears in the listing grid, and the system sends a setup email to the registered email address. 6) The customer designates specific outlets as Favourite (up to 5 outlets) to adjust the display order in the listing. 7) For first-time outlet setup, the customer provides the following additional information: a) Average Spend b) Visit Frequency c) Group Booking Size d) Restaurant Phone Number e) Type of Restaurant (optional) f) Opening Hours 8) The customer completes the setup, logs into the customer portal, and is directed to the Dashboard. 9) Upon selecting a record, the system will display the following details for the corresponding outlet: a) First Section: The outlet's name, address, associated group names, status, and the primary phone number linked to the outlet. b) Second Section: A listing grid of all users associated with the outlet, displaying the following information for each user: • Name • Email Address • Phone Number • Designation • Permissions • Status c) Third Section: A comprehensive overview of the campaign information related to the outlet, including details such as Total Campaign, Amount Spent, Total Bookings, and ROI etc. 10) The customer may edit any information on this screen by clicking the dedicated Edit button.

	11) A View Dashboard button will be provided. 12) When clicked, the system will redirect the customer to the Dashboard, displaying all relevant information specific to the selected outlet. 13) The customer can mark any outlet as Favorite (up to 5) by ping it. 14) The customer can click the Save button to store the information without making it available to use. 15) Incomplete Outlet Setup: If any required information is missing, the system prompts the customer to complete the necessary fields before saving. 16) Edit Existing Profile: The customer clicks on an existing outlet to update details. After making the necessary changes, the customer clicks Publish, and the system confirms the update. 17) When the customer selects the last option from the dropdown menu to add a new user, the system will redirect them to the Users Entry Form. 18) On this form, the customer can enter all required user details. 19) After publishing the form, the system will store the new user information and return the customer to the same entry form, where the newly added user will appear in the selection list.
Field Validations	1) Outlet Name: Textbox, alphanumeric, special characters supported, maximum length = 250 characters and mandatory. 2) Outlet Phone Number: Textbox with ISD dropdown, numeric, maximum length = 15 characters and mandatory. 3) Canari Phone Number: Drop down and mandatory. The options would be visible as per the existing subscription plan and as well as any additional phone number(s) that have been purchased. 4) Address Line 1: Textbox, alphanumeric, special characters supported, maximum length = 250 characters and mandatory. 5) Address Line 2: Textbox, alphanumeric, special characters supported, maximum length = 250 characters and mandatory. 6) City: Textbox, alphanumeric, special characters supported, maximum length = 250 characters and mandatory. 7) State/Province/Region: Textbox, alphanumeric, special characters supported, maximum length = 250 characters and mandatory. 8) ZIP/Postal Code: Textbox, alphanumeric, maximum length = 10 characters and mandatory. 9) Groups: Text area with comma-separated functionality, maximum length = 1000 characters and optional. 10) Average Spend: Textbox, numeric, should be > \$0 and mandatory. 11) Visit Frequency: Textbox, alphanumeric, maximum length = 50 characters and mandatory. 12) Group Booking Size: Textbox, numeric, should be > \$0 and mandatory. 13) Type of Outlet: Textbox, alphanumeric, maximum length = 50 characters and optional. 14) Group(s): Text area with comma-separated functionality, maximum length = 1000 characters and optional.

	15) Status: Mandatory and drop-down field with the following options: a) Active: The system stores all the calls and processes the analytics. b) Inactive: The system has recorded all the calls and analytics before Inactive status. All the phone numbers associated with this outlet is available to use. The customer can activate this outlet whenever needed and the system will start calculating the analytics. c) Expired: The system has recorded all the calls and analytics before the Expired status. All the phone numbers associated with this outlet are now available to use and the customer won't be able to activate this outlet for analytics purposes. 16) Opening Hours: The outlet's opening time for each day will be captured by selecting checkboxes and drop-downs along with an option to add break time (if applicable). 17) Description: Text area, alphanumeric along with special character supported, maximum length = 1000 characters and optional.
Business Validations	1) By default, only the account owner can view and manage the Outlets section. 2) Favourite outlets must be displayed directly in the listing, followed by other outlets. 3) Any updates to the outlet details should trigger an audit log entry to ensure traceability. 4) For Any changes in other outlets, the system will notify the group owner (In application notification and an email). However, from the setting, the customer can opt out of this function.
Notes	1) Ensure that the Outlet's grid is searchable and filterable to help customers quickly locate specific profiles. 2) Notifications for initial setup should include instructions for account setup and password creation. 3) For enhanced usability, consider adding a tooltip or guide to explain the all key functionalities.

6.7.4 Groups

This section allows customers to view, create, and manage groups for outlets, offering insights into cumulative metrics and AI-generated summaries.	
Preconditions	1) The customer must be logged into the Customer Portal with permission to access the Groups section. 2) The system must store data related to outlets, inquiries, bookings, campaigns, and associated metrics for each group. 3) Only account owners should have the option to create new groups by default.
Post Conditions	1) New groups are successfully added and displayed on the Groups screen in a card view format. 2) Updated group details are reflected immediately after editing. 3) Customers can filter and search for specific group information.

Normal Flows	1) The customer logs into the Customer Portal and navigates to Settings > Groups. 2) The system displays all existing groups in a card view format, showing: a) Total Outlets (associated with the respective group) b) Total Inquiries along with value (cumulative value of all outlets associated with the respective group) c) Total Bookings along with value (cumulative value of all outlets associated with the respective group) d) Total Campaigns with cumulative value (cumulative value of all outlets associated with the respective group) e) AI-generated summary (text format) for each group 3) To add a new group: a) The customer clicks + Add New. b) An entry form appears, prompting the customer to fill in: • Group Name • Select Outlet (dropdown of available outlets) • Add User(s) (multi-select option) c) The customer clicks Publish, and the system displays a confirmation message. d) The new group is added to the screen. 4) The customer uses the Filter button to apply filters or the Search box to locate specific groups or information. 5) To edit an existing group: a) The customer selects and clicks on a group card. b) The system displays an editable form with pre-filled information. c) The customer makes the necessary changes and clicks Publish. d) The system displays a confirmation message and updates the group details. 6) The customer will have the option to store the data without publishing the information by clicking on the Save button. 7) No Matching Results: If no results match the applied filter or search query, the system displays a message: "No groups found for the selected criteria".
Field Validations	1) Group Name: Textbox, alphanumeric along with special characters supported, maximum length = 250 characters and mandatory. 2) Select Outlet(s): Dropdown with multi-select option (minimum 2) and mandatory. The options would be fetched as per active Outlet. 3) Add User(s): Dropdown with multi-select option and mandatory. The options would be fetched as per active Users. 4) Description: Text area, alphanumeric along with special character supported, maximum length = 1000 characters and optional.
Business Validations	1) Only account owners can create or edit groups by default. 2) Each group must be uniquely named to prevent duplication. 3) Changes made to groups should immediately be reflected in associated metrics and AI summaries.

Notes	1) Ensure that the card view supports scalability for accounts with multiple groups. 2) Provide tooltips for metrics and AI summaries to enhance usability.
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6.7.5 Canari Numbers

This section allows customers to manage their associated phone numbers, purchase additional minutes or numbers, and assign numbers to specific outlets or campaigns.	
Preconditions	1) The customer must be logged into the Customer Portal with account owner access. 2) The system must store and retrieve phone number details and associated metrics. 3) Available plans for purchasing minutes or numbers must be pre-configured in the Admin Portal.
Post Conditions	1) Customers can view and manage all phone numbers associated with their account. 2) New purchases for minutes or numbers are reflected in the system. 3) Phone numbers are assigned to outlets or campaigns, triggering associated analytics.
Normal Flows	1) The customer logs into the Customer Portal and navigates to Settings > Canari Numbers. 2) The system displays: a) Widgets: Available Numbers, Average Call Time/Outlet, and Average Call Time/Campaign. b) Listing Grid: Canari Number, Outlet, Campaign, Type, Used Minutes, and Status. 3) The customer clicks Buy Additional Numbers or Buy Additional Minutes: a) The system fetches available plans from the Admin Portal. b) Once the customer selects a plan and clicks the "Buy Now" button, the system will automatically process the payment in the backend according to the selected plan. c) A confirmation message will be displayed on the screen upon successful completion of the purchase. 4) The system will display the remaining minutes at the top, with a detailed breakdown of the default allocated minutes as per the existing subscription plan, as well as any additional minutes that have been purchased. 5) To assign a number: a) When the customer selects the Available button, the system will prompt a question: "Where would you like to allocate this Canari number?" b) The customer will be presented with two options: New Outlet or New Campaign. c) Based on the customer's selection, the system will redirect the customer to the corresponding entry form screen for further action.

Field Validations	1) Canari Number: Drop-down and mandatory. The options would be fetched as per the existing subscription plan and additional phone number purchase (if there are any). 2) Outlet: Drop-down and mandatory. The options would be fetched as per the Outlets (only active outlet) section. 3) Campaign: Drop-down and mandatory. The options would be fetched as per the Campaigns (only active campaign) section. 4) Used Minutes: Read-only field. The system will calculate this information as per the usage of all the outlets and campaigns. 5) Status: Read-only field with the following options: a) Allocated: The number has been assigned to any outlet or campaign. b) Available: The number can be assigned to any outlet or campaign.
Business Validations	1) By default, only account owners can access and manage phone numbers. 2) Purchases for minutes or numbers must align with available plans configured in the Admin Portal. 3) Updates to phone numbers must trigger relevant analytics based on the new assignments.
Notes	1) Ensure that the widgets update dynamically to reflect the latest metrics. 2) Provide tooltips or help icons for each widget to explain the displayed metrics.

6.7.6 Billing

This section provides the customer with a detailed view of their invoices, enabling them to download the invoices or upgrade the plan.	
Preconditions	1) The customer must be logged into the Customer Portal with permission to access the Billings section. 2) Integration with Stripe must be configured for payment and invoice processing.
Post Conditions	1) The customer can view all billing history. 2) The customer can download invoices for their records. 3) The customer can upgrade the subscription plan.
Normal Flows	1) The customer logs into the Customer Portal and navigates to Settings > Billing. 2) The system displays a listing screen with columns: a) Invoice Number b) Invoice Date c) Invoice Amount d) Type (Subscription/Additional Phone Number) e) Paid Date 3) The customer can upgrade the subscription plan by clicking the dedicated button. 4) Upon clicking the View button, the customer will be redirected to the Stripe page, where the system will display additional details for

	the selected invoice. The customer will also have the option to download the invoice from this page. 5) Failed Payment: If the Stripe payment fails (while upgrading the subscription plan), the system displays an error message and provides an option to retry. 6) No Invoices Available: If no billing data is found for the customer, the system displays a message: "No invoices available".
Field Validation	NA
Business Validations	1) By default, only the account owner will have permission to this section (Billings). 2) Customers can only view invoices linked to their account. 3) Stripe payment confirmation must trigger an automatic status update in the system to Paid. 4) The system must send reminder emails based on the subscription plan type: a) Monthly Plans: Reminders sent 7 days, 3 days, and 1 day prior to expiry. Post-expiry, an additional email notifies the customer of the expiration. b) Yearly Plans: Reminders sent 1 month, 7 days, 3 days, and 1 day prior to expiry. Post-expiry, an additional email notifies the customer of the expiration. c) All reminders must include a payment link for renewal. 5) The customer can log in to the portal post-expiry; however: All screen details will appear blurred. A renewal payment link will be prominently displayed. 6) Alongside email reminders, the system will display reminder messages within the application.
Notes	1) Ensure secure data transmission between the Customer Portal and Stripe to protect sensitive payment information. 2) Include a tooltip or help guide explaining each column and action in the Billings section.

7. Non-Functional Requirement

7.1 Performance

- a) **Response Time:** The system should respond to user actions (e.g., navigation, form submission etc.) within **3 seconds** to maintain a seamless user experience.
- b) **Data Processing:** Updates to modules to be completed within **3 seconds** to ensure real-time accuracy.
- c) **Email Notifications:** System-generated emails (e.g., meeting details, registration etc.) to be sent within **1 minute** of the triggering the respective action.

Refer to the infrastructure details in Section 17 for server configuration and resource provisioning guidelines.

7.2 Scalability

- a) The system should be capable of supporting up to **100** concurrent users without performance degradation.
- b) The system architecture should support integration with additional third-party tools and databases as the application grows and business/customer requirements evolve. Refer to Section 17 for infrastructure expansion guidelines.

7.3 Reliability

- a) The system must ensure **99.9%** uptime, ensuring minimal downtime to provide consistent availability. Refer to Section 17 for server redundancy and failover configurations.

7.4 Usability

- a) The system should feature an intuitive, user-friendly interface with clear navigation paths.
- b) Tooltip guidance, error prompts, and feedback messages should be provided to enhance user experience and reduce the need for support intervention.

7.5 Security

- a) **Data Encryption:** All customer data should be encrypted at rest and in transit, meeting GDPR and regional data protection standards. Refer to Section 17 for infrastructure-level security configurations (e.g., TLS, encrypted storage).

7.6 Maintainability

- a) The system should be designed with modular code to facilitate seamless updates, allowing the development team to implement changes with minimal downtime.
- b) System logs related to user activity, error events, and changes in configurations should be retained for at least **6** months to assist in maintenance, troubleshooting, and auditing.

7.7 Compatibility

- a) The system should be compatible across major web browsers (e.g., Chrome, Firefox, Safari, Edge etc.).
- b) Regular compatibility testing should be conducted to ensure continued seamless integration with third-party services (e.g., Stripe, Calendly etc.).

8. System Requirement

8.1 User Interface

- **Programming Language:** Utilise JavaScript as the primary language for frontend web development to enable dynamic and interactive user interfaces.
- **Frontend Frameworks:** Employ React.js, a popular JavaScript library, for building user interfaces with a component-based architecture and efficient rendering using a virtual DOM.
- **Frontend Libraries:** Implement Redux or the Context API for state management across the application, ensuring predictable data flow and easier debugging. Additionally, use React Router

to handle navigation and routing within the single-page application (SPA), providing a seamless user experience.

8.2 Database

- **DBMS:** Utilise relational databases like PostgreSQL or MySQL for structured data (user accounts, subscriptions, settings).
- **Data Storage and Management:** Store user profiles, campaign information, billing details, and system configurations (Structured Data). Manage large volumes of call recordings and transcriptions efficiently (Unstructured Data).
- **Scalability and Performance:** Design the database architecture to scale horizontally to handle increasing data volumes and user loads. Implement proper indexing and query optimisation techniques to ensure fast data retrieval.
- **Data Integrity and Security:** Ensure transactional integrity through adherence to ACID properties. Encrypt sensitive data at rest using industry-standard encryption methods.
- **Backup and Recovery:** Schedule automated backups with secure storage solutions. Establish procedures for data recovery in case of corruption or loss.
- **High Availability:** Use database replication strategies to provide redundancy and minimise downtime. Implement failover systems to maintain service continuity during hardware or software failures.
- **Data Retention Policies:** Align data retention with legal requirements and company policies, deleting data after specified periods (e.g., call data retained for 2-3 months post-account cancellation).

8.3 Communication/Security

- **Secure Communication Protocols:** Enforce HTTPS with TLS 1.2 or higher for all web traffic to encrypt data in transit. Use secure API endpoints with proper authentication and encryption.
- **Authentication and Access Control:** Store passwords using strong hashing algorithms. Implement secure session handling with timeouts and protection against session hijacking.
- **Data Encryption and Protection:** Encrypt sensitive data in databases and storage systems. Collect only necessary data and avoid storing sensitive information like payment card details.
- **Network Security:** Deploy network firewalls to protect servers from unauthorised access.
- **Incident Response and Logging:** Develop procedures for detecting, responding to, and recovering from security incidents. Maintain detailed logs of system access and user activities for auditing purposes.
- **Compliance with Regulations:** Ensure data handling practices comply with data protection regulations in target regions. Obtain and manage user consent for data processing activities.
- **Third-Party Security:** Securely store and manage API keys and credentials for external services. Evaluate third-party service providers for their security practices.

8.4 System Integration

- **Integration with Call Providers:** Utilise APIs from call providers (Telnyx) for call connection, recording, transcription, and number management. Ensure integrations can handle increased call volumes as the user base grows.
- **Payment Gateway Integration:** Implement secure payment processing using Stripe's APIs for subscriptions and billing. Use Webhooks to receive real-time updates on payment statuses.

- **Scheduling Tool Integration:** Embed Calendly for demo bookings and synchronised scheduling data.
- **Data Synchronisation:** Ensure data consistency between the platform and external systems through reliable synchronisation methods. Implement robust error handling and retry mechanisms for failed integrations.
- **Security in Integration:** Use secure communication protocols when interacting with external APIs.

8.5 Constraints and Limitations

- **Initial Language Support:** The platform supports **English** only.
- **Device Compatibility:** The admin portal is optimised for desktop use.
- **Dependency on Third-Party Services:** Reliance on external providers (e.g., call providers, payment gateways) means that their availability and performance impact the platform. Updates or changes in third-party APIs may require platform adjustments.

8.6 Legal

8.6.1 Data Protection and Privacy

- **GDPR Compliance:** Implement measures to comply with the EU's GDPR, including data subject rights and data processing agreements.

8.6.2 Call Recording Regulations

- **Consent Laws:** Comply with regional laws regarding call recording consent (one-party vs. two-party consent).
- **Disclosure Requirements:** Provide necessary call recording disclosures (IVR) to callers as per legal requirements.

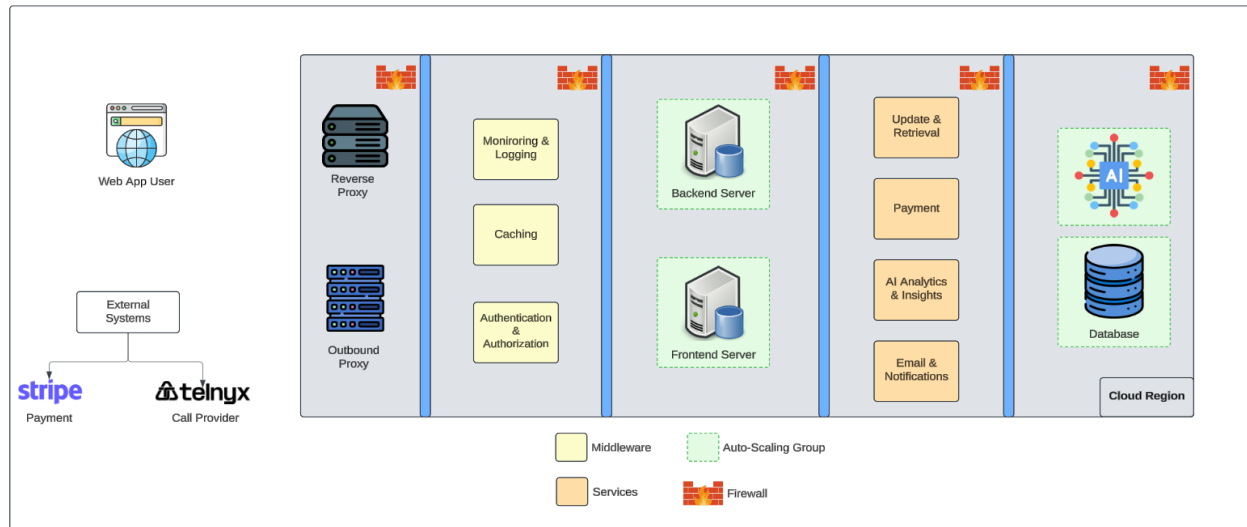
8.6.3 Liability Limitation

- **Disclaimers:** Include disclaimers limiting the platform's liability for data loss, service outages, or inaccuracies in analytics.

9. System Architecture

9.1 Overview

The Canari platform is architected to be a scalable, secure, and high-performance solution that meets the needs of hospitality brands seeking actionable insights from inbound call analytics. The system employs a modern N-tier architecture, leveraging cloud technologies and integrating with various external services to provide advanced features without compromising on security or performance.



9.2 N-tier Architecture

The platform is designed using an N-tier architecture, which divides the application into distinct components, each responsible for specific aspects of the system's functionality. This separation of concerns enhances scalability, maintainability, and security. The following are the components of the N-tier architecture:

9.2.1 Presentation Component

The Presentation Component is responsible for delivering a user-friendly interface to the customers and administrators, facilitating interaction with the platform's features. The following are the technologies and features that will be used at the time of development:

- **Frontend Frameworks:** React.js or Angular for building dynamic and responsive interfaces.
- **UI Libraries:** Consistent styling and component reuse.
- **Interactive Dashboards:** Visual representations of analytics data widgets.
- **Easy Navigation:** Clear menus to guide users through the platform.
- **Others:** Real-time validation, notifications, and confirmation dialogues to enhance user interaction.

9.2.2 Business Logic Component

The Business Logic Component contains the core functionality that processes data, enforces business rules, and orchestrates the application's operations. The following are the technologies and features that will be used at the time of development:

- **Backend Frameworks:** Python with Django for building scalable server-side applications.
- **Programming Languages:** JavaScript/TypeScript for Node.js or Python for Django, chosen for their ecosystem and performance.
- **Scalability:** Designed to handle increasing loads by scaling horizontally (adding more servers) or vertically (adding more resources).
- **Extensibility:** Modular structure allows for easy addition of new features or services.

- **Performance Optimisation:** Implements caching strategies and efficient algorithms to minimise latency.

9.2.3 Data Access Component

The Data Access Component manages communication between the application and the data storage systems, providing a consistent interface for data retrieval, manipulation, and storage. The following are the technologies and features that will be used during development:

- **Database Management Systems (DBMS):** Utilise relational databases like PostgreSQL for structured data (user accounts, subscriptions, settings) and NoSQL databases for unstructured data (logs).
- **Object-Relational Mapping (ORM):** Use Django ORM to abstract database interactions and map database tables to application objects, simplifying data manipulation and ensuring consistency.
- **Data Access Patterns:** Implement repository or data mapper patterns to decouple business logic from data access code, promoting maintainability and scalability.
- **Data Integrity and Security:** Ensure transactional integrity through adherence to ACID properties. Encrypt sensitive data at rest using industry-standard encryption methods, and in transit using SSL/TLS protocols.
- **Data Retention Policies:** Align data retention with legal requirements and company policies, deleting or archiving data after specified periods (e.g., call data retained for 2-3 months post-account cancellation).

9.2.4 Communication/Security Component

Ensuring secure communication and safeguarding user data are paramount for maintaining trust and complying with legal obligations. The platform implements comprehensive security measures across all communication channels and data-handling processes. The following are the technologies and features that will be used at the time of development:

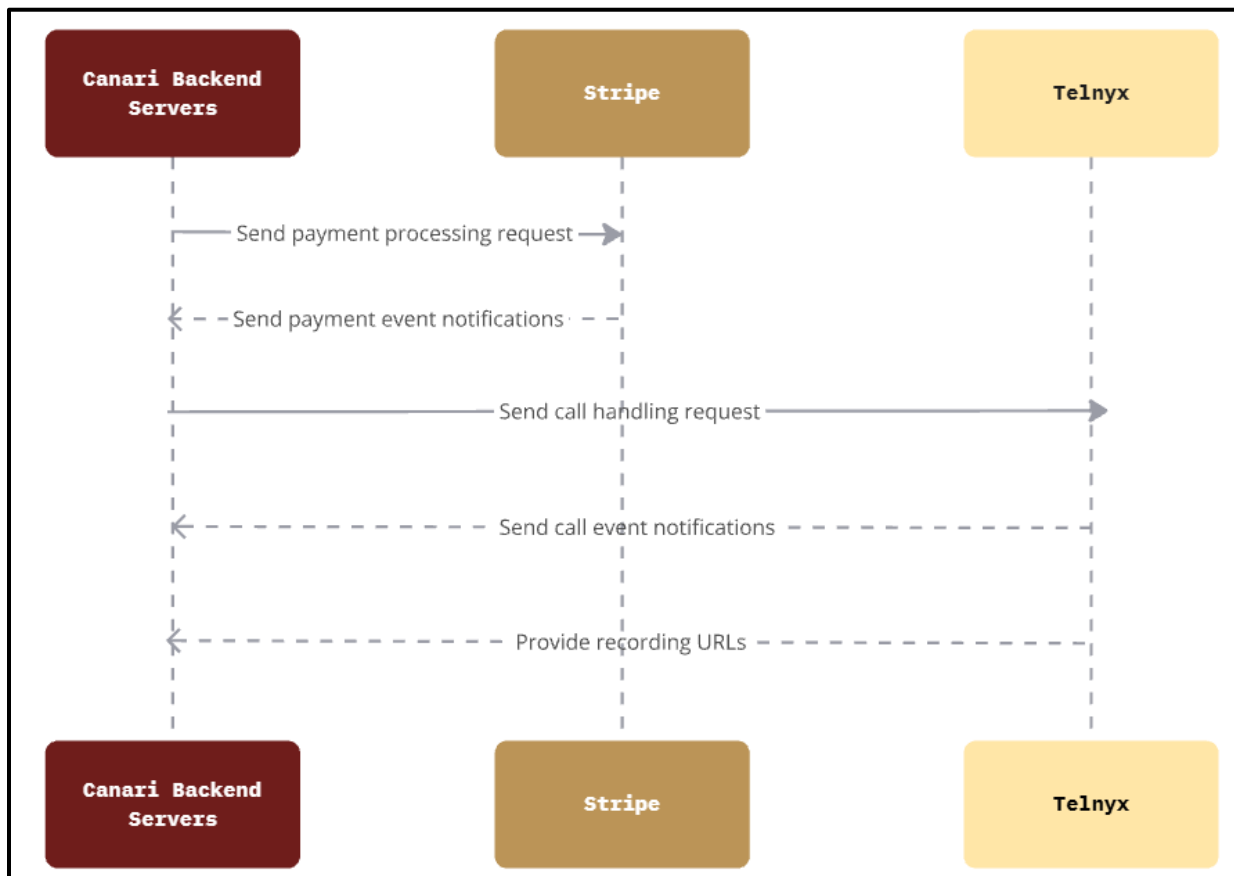
- **Transport Layer Security (TLS):** Implement SSL/TLS encryption for all data in transit to protect against eavesdropping and man-in-the-middle attacks, ensuring secure communication channels between client applications and backend services.
- **Authentication and Authorisation Frameworks:** Utilise JSON Web Tokens (JWT) for stateless authentication and authorisation, securely transmitting user credentials and permissions in each request to manage access control effectively.

9.2.5 API Component

The API Component serves as the communication bridge between the client-side applications (Presentation Component) and the server-side services (Business Logic Component). It exposes the platform's functionalities through a set of well-defined, secure, and versioned APIs, facilitating interaction with both internal components and external services while maintaining data integrity and security. In addition to handling requests from the Presentation Component, the API Component is responsible for integrating with external services critical to the platform's operations, such as payment gateway (Stripe), call provider (Telnyx), AI services for transcription

and sentiment analysis, email delivery services, and scheduling tools. The following are the technologies and features that will be used at the time of development:

- **External SDKs and Libraries:**
 - **Stripe:** For seamless integration with payment services.
 - **Telnyx:** For call management functionalities.
 - **AI (Optional):** For interacting with services like Google Cloud, AWS, or IBM Watson.
 - **Email Service:** For sending emails via SendGrid, Mailgun, or similar services.



9.3 Microservices Architecture

Instead of a monolithic application where presentation, business logic, data access, and integrations reside in a single codebase, the platform will be decomposed into smaller, loosely coupled microservices. Each microservice will own a specific domain or feature set, communicate primarily through well-defined APIs (often using REST or gRPC), and maintain its own data storage as appropriate. This ensures that changes in one microservice can be implemented, tested, deployed, and scaled independently, providing greater agility and fault tolerance.

9.3.1 Key Benefits

- **Scalability:** Each service can scale horizontally based on its own load and performance characteristics.
- **Maintainability:** Smaller, focused codebases are easier to understand, modify, and test.
- **Resilience:** Failures in one service are contained and do not necessarily bring down the entire platform.
- **Technology Flexibility:** Different microservices can use the most appropriate technology stacks, programming languages, and data stores for their specific domain.

9.3.2 Core Microservices and Their Responsibilities

- **User Management Service**

Responsibilities

- Handles user profiles, authentication, authorisation, and RBAC (Role-Based Access Control).

Features

- Account creation, login, password reset, and profile updates
- JWT issuance and validation for secure authentication
- Enforcing access controls and permissions

- **Customer & Outlet Management Service**

Responsibilities

- Manages customer accounts, **outlets**, groups, and hierarchical business structures.

Features

- CRUD operations for customer profiles, outlets, and groups
- Tracking subscription associations per customer
- Managing the linkages between customers, their outlets, and Canari phone numbers

- **Subscription & Billing Service**

Responsibilities

- Orchestrates subscriptions, payments, invoicing, and recurring billing using Stripe.

Features

- Managing subscription plans, upgrades, downgrades
- Generating and sending invoices via Stripe integration
- Handling one-time purchases (additional numbers, minutes)

- **Call Management & Telephony Service**

Responsibilities

- Integrates with Telnyx (or similar provider) for call routing, call recording, transcription requests, and capturing call metadata.

Features

- Provisioning and releasing virtual phone numbers (Canari numbers)
- Initiating and terminating calls, forwarding to outlets
- Storing and retrieving call recordings and call details records (CDRs)

- **Analytics & AI Service**

Responsibilities

- Provides call transcription, sentiment analysis, entity extraction, intent detection, and other AI-driven insights.

Features

- Integrating with external AI/NLP APIs
- Preprocessing call recordings, sending them for transcription
- Storing structured analytics data (call reasons, sentiment scores)
- Providing an API for dashboards and reporting

- **Campaign & Events Management Service**

Responsibilities

- Manages marketing campaigns, events, associated call tracking, and ROI calculations.

Features

- CRUD for campaigns and events
 - Linking campaigns to Canari numbers
 - Calculating performance metrics (bookings, inquiries, missed calls, potential revenue, ROI)
- Dashboard & Reporting Service

Responsibilities

- Aggregates data from multiple services (Analytics, Campaigns, Customers) to present a consolidated dashboard for admins and customers.

Features

- Generating real-time dashboards, charts, and reports
 - Providing filtered and customized views per customer/group/outlet/campaign
- Notifications & Communication Service

Responsibilities

- Manages system-generated emails, SMS, and in-app notifications to customers and admins.

Features

- Sending onboarding emails, payment reminders, password resets
 - Triggering notifications on subscription expiry, new campaign launched, missed calls, etc.

9.3.3 Communication and Integration Patterns

- **Service-to-Service Communication**
 - Primarily via RESTful APIs or gRPC endpoints over HTTPS for synchronous requests.
 - Use of an API Gateway to route external traffic to the appropriate microservices.
 - Internal services may also leverage asynchronous messaging (e.g., event bus or message queue) to reduce coupling and improve resilience.
- **Asynchronous Messaging**

- Event-driven architecture where services publish domain events (e.g., "PaymentCompleted", "CallRecorded", "CampaignCreated") to a message broker.
- Other services subscribe to relevant topics/events to update their records or trigger subsequent actions, enabling eventual consistency and decoupled workflows.

Security and Compliance

- Authentication & Authorisation
 - The User Management Service issues JWTs upon successful login.
 - All requests to other microservices must carry valid tokens.
 - Role-based and attribute-based access control enforced at the gateway or at each service boundary.
- Transport-Level Security
 - All communication between services and with external clients must be encrypted with TLS.
- Auditing & Logging
 - Centralized logging with correlation IDs to trace requests across multiple services.
 - Audit logs stored in a secure, immutable location for regulatory compliance

9.4 Method of Development

The development methodology focuses on delivering a high-quality product through iterative processes, continuous feedback, and adherence to best practices. The following are the methods/tools/practices that will be used at the time of development:

- **Agile: Scrum** framework to organise work into sprints, hold daily stand-ups, sprint planning, reviews, and retrospectives. This will encourage cross-functional team collaboration and frequent communication.
- **Version Control:** Git will be utilised for source code management. Implement automated testing and deployment using tools like Jenkins, GitLab CI/CD, or GitHub Actions.
- **Code Reviews:** Peer reviews to ensure code quality and adherence to standards.
- **Static Code Analysis:** Use tools like ESLint or Pylint to detect code issues early.
- **Testing Strategies:** Write tests for individual units of code using frameworks like Jest (JavaScript) or PyTest (Python) for unit testing. Verify that different components work together as expected for integration testing.
- **Documentation:** Comprehensive technical documentation regarding APIs, system architecture, and deployment processes will be maintained. A user manual will be drafted to provide resources for end-users.

9.5 High-Level Technical Design

Users interact with the platform through a web interface, which sends requests from the frontend to the backend API. The backend then processes these requests by applying business rules and interacting with databases through the Data Access Component for data storage and retrieval. Additionally, the Integration Component manages communication with external APIs to incorporate additional functionalities. After processing, the backend sends the appropriate responses back to the front end, ensuring that the information is effectively presented to the user.

Note: The technical design is subject to refinement based on evolving project requirements. The architecture aims to balance complexity with the need for scalability, security, and performance.

10. Acceptance Criteria

The Acceptance Criteria define the measurable conditions that the Website, Admin Portal, and Customer Portal must meet to ensure the system's functionality aligns with business requirements. These criteria serve as a benchmark for validating the system's usability, performance, and compliance with industry standards.

a) Website

- The website must be accessible globally, optimised for the targeted regions (UK, USA, UAE), and responsive across all devices (desktop, tablet, and mobile).
- Users must be able to Book a Demo seamlessly via Calendly integration, with confirmation emails sent to both the customer and the admin team.
- Existing customers must be able to access the Login button to navigate to the Customer Portal.
- SEO compliance and Google Analytics tracking must be implemented for all pages to monitor site traffic and user engagement.
- All forms must include field validation and display appropriate error messages for incomplete or invalid submissions.

b) Admin Portal

- Admin users must be able to manage all modules with role-based access control implemented.
- Dashboards must provide accurate analytics with dynamic filtering.
- Integration with third-party tools (e.g., Stripe for payments, Calendly for scheduling) must function as expected.
- Audit logs must capture all admin actions, ensuring transparency and traceability.
- The system must ensure secure data handling, complying with GDPR and other regional data protection regulations.

c) Customer Portal

- Customers must have seamless access to manage their account details with updates reflected system-wide.
- AI Search functionality must respond accurately to user queries, offering both text and visual analytics, and incorporate feedback for continuous improvement.
- Dashboards must be intuitive, displaying tailored analytics and Stamps for outlets, campaigns, and customer engagement.

- Permission-based access must ensure that only account owners can access and modify restricted sections (e.g., Canari Numbers, Groups).
- Payment processing for buying additional number(s) or minutes must be secure, with confirmation emails generated post-transaction.
- Multilingual and accessibility features must be scalable, ensuring smooth integration for additional languages or regions in the future.

11. Product Future

The product is designed to evolve and adapt to meet future market demands and technological advancements. The following outlines potential enhancements and expansions planned for the product:

- Region-Wise Plans:** Region-specific subscription plans may be introduced to enhance operational efficiency for both administrators and customers.
- Geographical Expansion:** Currently targeted at the UK, USA, and UAE markets, the product will be scalable to include additional regions based on customer demand and market trends. Regional configurations will include compliance with local regulations, cultural adaptations, and currency support.
- Advanced Authentication Methods:** Integration of Sign-in options with Apple, Google, and Facebook to enhance accessibility and streamline the login process for users. Future consideration for biometric authentication (e.g., fingerprint or facial recognition) to further enhance security.
- Multi-Language Support:** Expansion to support additional languages, such as Arabic, to cater to non-English-speaking users in diverse markets. Implementation of language preference settings for both Admin and Customer Portals.
- Enhanced AI Capabilities:** Introduction of audio and video summaries for dashboards and analytics, allowing users to consume insights in varied formats. AI-driven predictive analytics to provide actionable insights for campaigns and customer engagement strategies.
- Enhance Ask Canari:** In the future, the system may include the capability to generate visual elements, such as graphs and charts, based on user queries for the Ask Canari module. Additionally, voice command functionalities could be integrated to enhance user interaction and improve overall system accessibility.
- Live Chatbot:** To further enhance the user experience and ensure seamless customer support, the integration of a live AI-powered chatbot within the customer portal can be introduced. This chatbot will provide immediate, automated assistance to customers, addressing common queries with contextually relevant responses. In cases where the AI-driven chatbot is unable to resolve the issue to the customer's satisfaction, the system will automatically escalate the query to a human support executive. This escalation process ensures that complex or unresolved issues are promptly addressed by a qualified professional, thereby improving the overall support functionality and enhancing customer satisfaction.
- Mobile App:** A dedicated mobile app for iOS and Android platforms may be introduced to provide a more seamless and intuitive user experience.
- Integration with External Platforms:** Seamless integration with CRM tools, marketing platforms, and other third-party applications to extend functionality.

12. Technical Evaluation of Call Providers

The core use case involves customers calling a designated number provided by a call provider, which then redirects the call to the client's number. During this call, the call provider must remain in the call path to enable call recording. The primary focus is on simplicity and ease of use, allowing end-users to seamlessly acquire phone numbers and utilise call recording features through the platform without direct interaction with the underlying telephony provider. The selected vendor is **Telnyx**.

12.1 Functionalities

- a) Number Allocation: Ability to programmatically purchase and allocate phone numbers to end-users via API.
- b) Call Connection: Incoming calls to the provider's number should be connected to the client's fixed number, with the provider remaining in the call path for recording.
- c) Call Recording: Automatic recording of calls with access to recordings and metadata (call duration, caller ID, etc.).
- d) API Integration: Seamless integration with the provider's APIs for number management, call handling, and recording retrieval.
- e) Reselling and White-labeling: The platform should handle all interactions with the telephony provider, allowing reselling under the platform's branding without end-users needing direct accounts with the provider.

12.2 Pricing

Cost Calculation for 50 Customers with 1 phone and 3 campaigns each. 5 of these customers will have 10 campaigns each instead:

- a) Customers: **50**
- b) Phone Numbers per Customer:
 - 45 Customers: 1 main number + 3 campaign numbers = **4 numbers each**
 - 5 Customers: 1 main number + 10 campaign numbers = **11 numbers each**
- c) Total Phone Numbers: **235 numbers**
- d) Average Call Duration: **3 minutes**
- e) Total Calls per Week: **50 calls**
- f) Rates for **UK** geographical numbers:
 - Non-Recurring Charge (NRC) per Number: \$1.00
 - Monthly Recurring Charge (MRC) per Number: \$1.00
 - Inbound Call Rate: \$0.0055 per minute
 - Outbound Call Rate (Call Forwarding): \$0.0070 per minute
 - Call Recording Rate (Origination and Termination): \$0.0016 per minute each (total \$0.0032 per minute)
 - Call Control Rate (Origination and Termination): \$0.0016 per minute each (total \$0.0032 per minute)
- g) Costs for **UK** geographical numbers:
 - Weekly Total Cost: **\$61.59**
 - Monthly Total Cost: **\$246.34**

- Yearly Cost (Recurring Charges Only): **\$2,967.42**
- One-Time Costs (NRC): **\$235.00**
- Total First-Year Cost: $\$2,967.42 + \$235.00 = \$3,202.42$

13. AI Requirement

The canari platform integrates a comprehensive suite of Artificial Intelligence (AI) features designed to transform inbound call data into actionable insights for hospitality brands. Leveraging advanced machine learning and natural language processing (NLP) techniques, these features enhance customer engagement, optimise marketing strategies, and improve operational efficiency. Below is an overview of each AI feature, including the proposed models to be utilised and potential paid AI solutions where applicable.

13.1 Intent Detection

Intent Detection identifies the primary purpose behind each inbound call, such as making a reservation, placing an order, or lodging a complaint. Accurate intent classification enables streamlined response strategies and efficient resource allocation.

a) **Proposed Models:**

- BERT (Bidirectional Encoder Representations from Transformers): Renowned for its strong contextual understanding, BERT effectively captures the nuanced meanings in conversations.
- RoBERTa (Robustly Optimised BERT Pretraining Approach): An optimised variant of BERT that offers improved performance on various NLP tasks.

b) **Paid AI Solutions:**

- OpenAI GPT-4 API: Offers high accuracy in intent detection with customisable fine-tuning options.
- Estimated Cost: Approximately \$0.03 per 1,000 tokens processed. To make it simple, for each 3-minute call you process through the OpenAI GPT-4 API, it will cost about 1.7 cents. This estimate is based on an average speaking pace, where a typical 3-minute conversation includes around 420 spoken words. When transcribed, this amounts to roughly 560 tokens (the units used for billing by the API). Since the API charges approximately \$0.03 for every 1,000 tokens processed, each call would cost:
 - Estimated Cost per 3-minute call (on average) = 0.8 cents

13.2 Emotion Recognition

Emotion Recognition analyses the emotional tone of callers, detecting specific emotions like frustration, happiness, anger, or confusion. This feature provides a deeper understanding of customer satisfaction and engagement levels.

a) **Proposed Models:**

- EmoBERT: A specialised version of BERT fine-tuned for emotion detection tasks.
- LSTM (Long Short-Term Memory) Networks: Effective for capturing temporal dependencies in speech data to identify emotions over time.

b) **Paid AI Solutions:**

- IBM Watson Tone Analyser: Provides comprehensive emotion detection capabilities.

- Estimated Cost: Starting at \$0.0025 per API call, with packages available based on usage volume.
- Estimated Cost per 3-minute call (on average) = 0.25 cents

13.3 Call Reason Categorisation

Call Reason Categorisation classifies calls into predefined categories such as reservations, complaints, inquiries, and feedback. This systematic classification aids in organising call data for efficient analysis and reporting.

a) **Proposed Models:**

- FastText: A lightweight and efficient model suitable for high-throughput text classification tasks.
- Support Vector Machines (SVM): Known for their effectiveness in high-dimensional spaces, particularly for clear margin separations in categorisation tasks.

b) **Paid AI Solutions:**

- Google Cloud AutoML Natural Language: Facilitates custom classification models with minimal training data.
- Estimated Cost: Approximately \$1.00 per 1,000 text records.
- Estimated Cost per 3-minute call (on average) = 0.1 cents

13.4 Customer Satisfaction Prediction

Predicts customer satisfaction levels based on conversation content and tone using machine learning models. This predictive capability allows businesses to proactively address customer concerns and improve service quality.

a) **Proposed Models:**

- Gradient Boosting Machines (e.g., XGBoost): Highly effective for regression and classification tasks with structured data.
- Neural Networks: Capable of capturing complex patterns in conversation data to predict satisfaction levels accurately.

b) **Paid AI Solutions:**

- Azure Machine Learning: Offers scalable machine learning services with support for various models.
- Estimated Cost: Starting at \$0.10 per hour for compute resources, plus storage costs.
- Estimated Cost per 3-minute call (on average) = 0.0083 cents

13.5 Average Call Duration

Measures the length of each call to analyse efficiency and customer engagement. Tracking call duration helps in assessing the effectiveness of interactions and optimising call handling processes.

a) **Proposed Models:**

- Linear Regression: Simple and interpretable model suitable for predicting continuous variables like call duration.
- Time Series Forecasting Models (e.g., ARIMA): Useful for analysing and predicting trends in call durations over time.

b) Paid AI Solutions:

- Amazon Forecast: Provides scalable forecasting services for time-series data.
- Estimated Cost: Approximately \$0.02 per forecasted data point.
- Estimated Cost per 3-minute call (on average) = 0.002 cents

13.6 Sentiment Trajectory

Tracks how the sentiment of a conversation evolves throughout the call, providing deeper insights into customer experiences and interaction dynamics.

a) Proposed Models:

- Transformer-based Models (e.g., BERT for Sequence Classification): Capable of understanding sentiment changes over the course of the conversation.
- Hidden Markov Models (HMM): Effective for modelling sequential data and tracking sentiment transitions.

b) Paid AI Solutions:

- OpenAI GPT-4 API: Can be leveraged to analyse sentiment shifts in conversation sequences.
- Estimated Cost: Approximately \$0.03 per 1,000 tokens processed.
- Estimated Cost per 3-minute call (on average) = 0.8 cents

13.7 Call Transcription

Converts audio recordings of calls into text, enabling easier analysis, record-keeping, and searchability. Transcriptions serve as the foundation for other AI features like intent detection and sentiment analysis.

a) Proposed Models:

- Google Cloud Speech-to-Text API: Offers high accuracy and supports multiple languages, suitable for reliable transcription.
- Mozilla DeepSpeech: An open-source alternative leveraging deep learning for speech recognition tasks.

b) Paid AI Solutions:

- Google Cloud Speech-to-Text API
 - Estimated Cost: Approximately \$0.006 per 15 seconds of audio processed.
 - Estimated Cost per 3-minute call (on average) = 7.2 cents
- Rev.ai: Provides high-quality transcription services with fast turnaround times.
 - Estimated Cost: Approximately \$0.25 per minute of audio.
 - Estimated Cost per 3-minute call (on average) = 75 cents

13.8 Time Series and Trend Analysis

Analyses call data over time to identify trends, patterns, and peak periods. Understanding temporal variations aids in strategic planning and resource allocation.

a) Proposed Models:

- ARIMA (AutoRegressive Integrated Moving Average): Traditional model effective for univariate time series forecasting.

- Prophet (by Facebook): Robust to missing data and shifts in trends, suitable for business forecasting.

b) **Paid AI Solutions:**

- Facebook Prophet via Azure ML: Offers integrated time-series forecasting with scalability.
- Estimated Cost: Included within Azure Machine Learning pricing (starting at \$0.10 per hour for compute resources).
- Estimated Cost per 3-minute call (on average) = 0.000833 cents

13.9 Gender Detection

Identifies the gender of the caller based on voice analysis and speech patterns, allowing for personalised service and tailored marketing strategies.

a) **Proposed Models:**

- Convolutional Neural Networks (CNNs) for Audio Classification: Effective in distinguishing gender-related features in voice data.
- Support Vector Machines (SVM) with Feature Engineering: Utilises engineered audio features for gender classification tasks.

b) **Paid AI Solutions:**

- Microsoft Azure Speaker Recognition: Provides gender detection capabilities as part of its speaker recognition services.
- Estimated Cost: Approximately \$1 per 1,000 audio segments processed.
- Estimated Cost per 3-minute call (on average) = 0.1 cents

13.10 Entity Extraction

Extracts key entities from conversations, such as names, dates, menu items, or locations, providing structured data for actionable insights and reporting.

a) **Proposed Models:**

- spaCy's Named Entity Recognition (NER): Efficient and accurate for extracting predefined entities from text.
- BERT-based NER Models: Leveraging transformers for context-aware entity extraction.

b) **Paid AI Solutions:**

- Amazon Comprehend: Offers robust entity extraction services with support for custom entity recognition.
 - Estimated Cost: Approximately \$0.0001 per unit (100 characters).
 - Estimated Cost per 3-minute call (on average) = 0.234 cents
- Google Cloud Natural Language API: Provides entity extraction with high accuracy and scalability.
 - Estimated Cost: Approximately \$1.00 per 1,000 units.
 - Estimated Cost per 3-minute call (on average) = 0.234 cents

13.11 Sentiment Analysis

Determines the overall sentiment of the call as positive, negative, or neutral, offering a quick assessment of customer feelings and satisfaction levels during interactions.

a) **Proposed Models:**

- VADER (Valence Aware Dictionary and sEntiment Reasoner): Rule-based model effective for sentiment analysis in social media and conversational text.
- BERT for Sentiment Classification: Provides deep contextual understanding, improving sentiment detection accuracy.

b) Paid AI Solutions:

- OpenAI GPT-4 API: Can be utilised for sophisticated sentiment analysis tasks.
 - Estimated Cost: Approximately \$0.03 per 1,000 tokens processed.
 - Estimated Cost per 3-minute call (on average) = 0.8 cents
- IBM Watson Natural Language Understanding: Offers sentiment analysis with customisable parameters.
 - Estimated Cost: Approximately \$0.003 per unit (1,000 characters).
 - Estimated Cost per 3-minute call (on average) = 0.702 cents

13.12 Demographic Inference - Gender, Tourist vs Local

Infers demographic attributes of callers, including gender, and whether the caller is a tourist or a local. This information enables businesses to personalize interactions and tailor marketing strategies based on customer demographics.

a) Proposed Models:

- Deep Learning Models (e.g., CNNs for Audio Analysis): Analyse voice characteristics to infer gender.
- Classification Algorithms (e.g., Random Forests): Use call metadata and speech patterns to determine if the caller is a tourist or a local.

b) Paid AI Solutions:

- Microsoft Azure Cognitive Services - Speaker Recognition: Provides demographic inference capabilities.
 - Estimated Cost: Approximately \$1 per 1,000 audio segments processed.
 - Estimated Cost per 3-minute call (on average) = 0.1 cents
- Amazon Rekognition: Can infer demographics from audio and video data.
 - Estimated Cost: Approximately \$0.001 per second of audio analysed.
 - Estimated Cost per 3-minute call (on average) = 18 cents

13.13 Ask Canari

An AI-powered search bar that allows administrators and users to query the platform for specific information, reports, or insights. This feature facilitates quick access to data without navigating through multiple menus.

a) Proposed Models:

- Elasticsearch with NLP Enhancements: Provides fast and relevant search results based on natural language queries.
- GPT-based Models (e.g., OpenAI's GPT-4): Enables more conversational and context-aware search interactions.

b) Paid AI Solutions:

- OpenAI GPT-4 API: Enhances search capabilities with conversational AI.
 - Estimated Cost: Approximately \$0.03 per 1,000 tokens processed.

- Estimated Cost per 3-minute call (on average) = 4.5 cents
- Elastic Cloud (Elasticsearch Service): Offers managed search and analytics.
 - Estimated Cost: Starting at approximately \$16 per month for basic tiers, scaling with usage.
 - Estimated Cost per 3-minute call (on average) = 0.16 cents

13.14 New vs Returning Customer - Phrases Said and Recognising Phone Number

Identifies whether a caller is a new or returning customer by analysing phrases used during the call and recognising phone numbers. This differentiation helps in understanding customer loyalty and tailoring interactions accordingly.

a) **Proposed Models:**

- Natural Language Processing (NLP) Models: Analyse call transcripts to detect phrases indicative of new or returning customers.
- Phone Number Recognition Systems: Match caller phone numbers against existing customer databases to identify returning customers.

b) **Paid AI Solutions:**

- OpenAI GPT-4 API: Assists in analysing conversational phrases to determine customer status.
 - Estimated Cost: Approximately \$0.03 per 1,000 tokens processed.
 - Estimated Cost per 3-minute call (on average) = 1.8 cents

13.15 Lost Opportunities

Identifies the reasons behind lost opportunities, such as technical issues, dissatisfaction, or external factors. Understanding why calls are terminated prematurely helps businesses address underlying problems and improve customer retention.

a) **Proposed Models:**

- Sequence Classification Models (e.g., LSTM, Transformers): Analyse the final segments of call transcripts to determine lost opportunities.
- Pattern Recognition Algorithms: Detect common patterns or keywords associated with call terminations.

b) **Paid AI Solutions:**

- OpenAI GPT-4 API: Facilitates analysis of conversation endings to identify lost opportunities.
 - Estimated Cost: Approximately \$0.03 per 1,000 tokens processed.
 - Estimated Cost per 3-minute call (on average) = 1.8 cents
- IBM Watson Natural Language Understanding: Can be configured to detect specific patterns leading to lost opportunities.
 - Estimated Cost: Approximately \$0.003 per unit (1,000 characters).
 - Estimated Cost per 3-minute call (on average) = 0.7 cents

13.16 Complaints Section

Categorises and analyses complaints received during calls to identify recurring issues and areas needing improvement. This feature enables businesses to systematically address customer grievances and enhance service quality.

a) **Proposed Models:**

- Text Classification Models (e.g., BERT, FastText): Classify complaints into specific categories based on content.
- Topic Modeling (e.g., LDA - Latent Dirichlet Allocation): Identify common themes and topics within complaints for comprehensive analysis.

b) **Paid AI Solutions:**

- Google Cloud Natural Language API: Offers robust text classification and topic modelling capabilities.
 - Estimated Cost: Approximately \$1.00 per 1,000 units.
 - Estimated Cost per 3-minute call (on average) = 0.234 cents
- Amazon Comprehend: Provides comprehensive complaint categorisation and topic modelling.
 - Estimated Cost: Approximately \$0.0001 per unit (100 characters).
 - Estimated Cost per 3-minute call (on average) = 0.234 cents

13.17 Peak Call Times & Projections

Analyses call data to identify peak call times and project future call volumes based on historical trends. This information aids in optimising staffing and resource allocation to handle high-demand periods efficiently.

a) **Proposed Models:**

- Time Series Forecasting Models (e.g., ARIMA, Prophet): Predict future call volumes based on past data.
- Machine Learning Regression Models (e.g., XGBoost): Forecast call trends considering multiple influencing factors such as holidays and promotions.

b) **Paid AI Solutions:**

- Facebook Prophet via Azure ML: Offers integrated time-series forecasting with scalability.
 - Estimated Cost: Included within Azure Machine Learning pricing (starting at \$0.10 per hour for compute resources).
 - Estimated Cost per 3-minute call (on average) = 0.03 cents.
- Amazon Forecast: Provides scalable forecasting services for time-series data.
 - Estimated Cost: Approximately \$0.02 per forecasted data point.
 - Estimated Cost per 3-minute call (on average) = 0.2 cents

14. Integration of Third-Party Data: Holiday Calendars by Country

To enhance the contextual relevance of analytics, the Canari platform integrates third-party data such as holiday calendars. This integration allows the platform to account for seasonal variations, special events, and holidays that may influence call volumes and customer behaviour.

14.1 Integration Approaches

a) **Holiday Calendar Feed:**

- Description: Automatically imports holiday data from reputable sources through API feeds.

- Advantages: Ensures up-to-date and accurate holiday information with minimal manual intervention.
 - Implementation: Utilise APIs from public holiday data providers to fetch and update holiday schedules regularly.
- b) Manual Entry:
- Description: Allows administrators to manually input holiday dates and relevant events through the admin portal.
 - Advantages: Provides flexibility to include custom or region-specific holidays not covered by standard feeds.
 - Implementation: Design user-friendly forms within the admin panel for easy addition and modification of holiday data.
- c) Hybrid Approach:
- Description: Combines automated feeds with manual overrides to ensure comprehensive and customisable holiday data integration.
 - Advantages: Balances accuracy and flexibility, allowing for both automated updates and manual adjustments.
 - Implementation: Enable automatic data import from holiday feeds while providing administrators the ability to add, edit, or delete entries as needed.

14.2 Admin Portal Integration

- a) Data Entry Interface: Develop intuitive interfaces within the admin portal for managing holiday calendars, supporting both automated imports and manual entries.
- b) Customisation Options: Allow administrators to specify regions, types of holidays, and any additional metadata that may be relevant for analytics (e.g., public vs. private holidays).
- c) Data Utilisation: Incorporate holiday data into time-series and trend analysis to contextualize call volume spikes or drops. Adjust sentiment and intent analysis models to account for holiday-related interactions and behaviors.

14.3 Benefits

- a) Enhanced Accuracy: Provides context to anomalies in call data, distinguishing between regular trends and holiday-induced variations.
- b) Strategic Planning: Enables businesses to anticipate and prepare for increased or decreased call volumes during holidays and special events.
- c) Improved Customer Insights: Helps in understanding customer behavior patterns in relation to holidays, informing marketing and service strategies.

14.4 Implementation Considerations

- a) Data Accuracy and Reliability: Ensure that holiday data sources are reliable and updated regularly to maintain the integrity of analytics.
- b) User Permissions: Restrict access to holiday calendar management features to authorised administrators to prevent unauthorised modifications.
- c) Integration Testing: Conduct thorough testing to ensure seamless integration of holiday data with existing analytics and reporting functionalities.

14.5 Model Flexibility and Adaptability

While the above models are selected based on their current effectiveness and suitability for each AI feature, it is essential to maintain flexibility throughout the development process. As the project progresses, the team will continuously evaluate model performance against real-world data and user feedback. Should alternative models demonstrate higher accuracy, efficiency, or better alignment with business objectives, the platform will adopt these superior approaches to ensure optimal performance and user satisfaction.

14.6 Conclusion

The integration of these AI features, supported by carefully selected machine learning and NLP models, positions the canari platform as a cutting-edge solution for hospitality brands. By transforming inbound call data into rich, actionable insights and incorporating contextual third-party data like holiday calendars, the platform empowers businesses to enhance customer experiences, streamline operations, and drive informed decision-making. The commitment to model adaptability ensures that canari remains at the forefront of AI-driven analytics, continuously improving to meet evolving industry needs.

15. Stripe Integration and Features

Stripe is integrated into the canari platform to handle all payment-related functionalities, including processing one-time and recurring payments, managing subscriptions, invoicing, and ensuring compliance with financial regulations. This integration enables canari to offer a reliable and secure billing system that caters to the needs of hospitality brands across the UAE, UK, and USA.

15.1 Key Objectives

- a) Secure Payment Processing: Ensure all transactions are handled securely, protecting sensitive financial data.
- b) Automated Billing: Streamline subscription management and billing cycles with minimal manual intervention.
- c) Scalability: Support a growing user base and increasing transaction volumes without compromising performance.
- d) Compliance: Adhere to global financial regulations and standards, including PCI DSS compliance.
- e) User Experience: Provide a smooth and intuitive payment experience for clients, enhancing satisfaction and retention.

15.2 Stripe Features Utilised

15.2.1 Payment Processing

Stripe facilitates the processing of both one-time and recurring payments through a secure and efficient infrastructure. It supports a wide range of payment methods, including credit and debit cards, digital wallets (e.g., Apple Pay, Google Pay), and bank transfers.

15.2.2 Subscription Management

Stripe's subscription management capabilities allow canari to offer various subscription plans to its clients, manage recurring billing cycles, and handle upgrades or downgrades seamlessly.

15.2.3 Invoicing

Stripe Invoicing allows canari to generate and send detailed invoices to clients, providing transparency and facilitating smooth financial transactions.

15.2.4 Webhooks

Stripe Webhooks enable real-time communication between Stripe and the canari platform, allowing the system to respond to various events such as payment successes, failures, and subscription changes.

15.2.5 Fraud Detection and Prevention

Stripe Radar is Stripe's built-in fraud detection system that leverages machine learning to identify and prevent fraudulent transactions, protecting both canari and its clients from financial losses.

15.2.6 PCI DSS Compliance

Stripe ensures that canari remains compliant with the Payment Card Industry Data Security Standard (PCI DSS), which is essential for handling credit and debit card information securely.

15.2.7 Financial Reporting and Analytics

Stripe offers detailed financial reporting and analytics tools that provide insights into transaction volumes, revenue streams, and financial health, enabling informed business decisions.

15.2.8 Customer Portal

Stripe Customer Portal allows clients to manage their own subscriptions, update payment methods, view billing history, and handle other account-related tasks independently.

15.3 Integration Strategy

- a) **API Integration:** The integration between canari and Stripe is primarily managed through Stripe's comprehensive APIs. This allows for seamless interaction between the platform and Stripe's services, enabling real-time processing and management of payments and subscriptions.
- b) **Webhook Configuration:** Stripe Webhooks are configured to notify the canari platform of important events, such as payment successes, failures, subscription renewals, and

chargebacks. These notifications trigger automated workflows within canari to update user access, send confirmation emails, or alert the support team as necessary.

- c) Secure Data Handling: All communication with Stripe is conducted over secure HTTPS connections. Sensitive data, such as API keys and webhook signatures, are stored securely using environment variables and encrypted storage solutions to prevent unauthorised access.
- d) User Experience Integration: Stripe's Checkout and Customer Portal are embedded within the canari platform to provide a seamless and branded user experience. This integration ensures that clients can manage their billing information and subscriptions without leaving the platform.
- e) Testing and Validation: Thorough testing is conducted in Stripe's sandbox environment to validate all payment flows, subscription management, and webhook handling before deploying to the production environment. Automated tests are implemented to ensure ongoing reliability and performance of the payment system.

15.4 Fee Structure (UK)

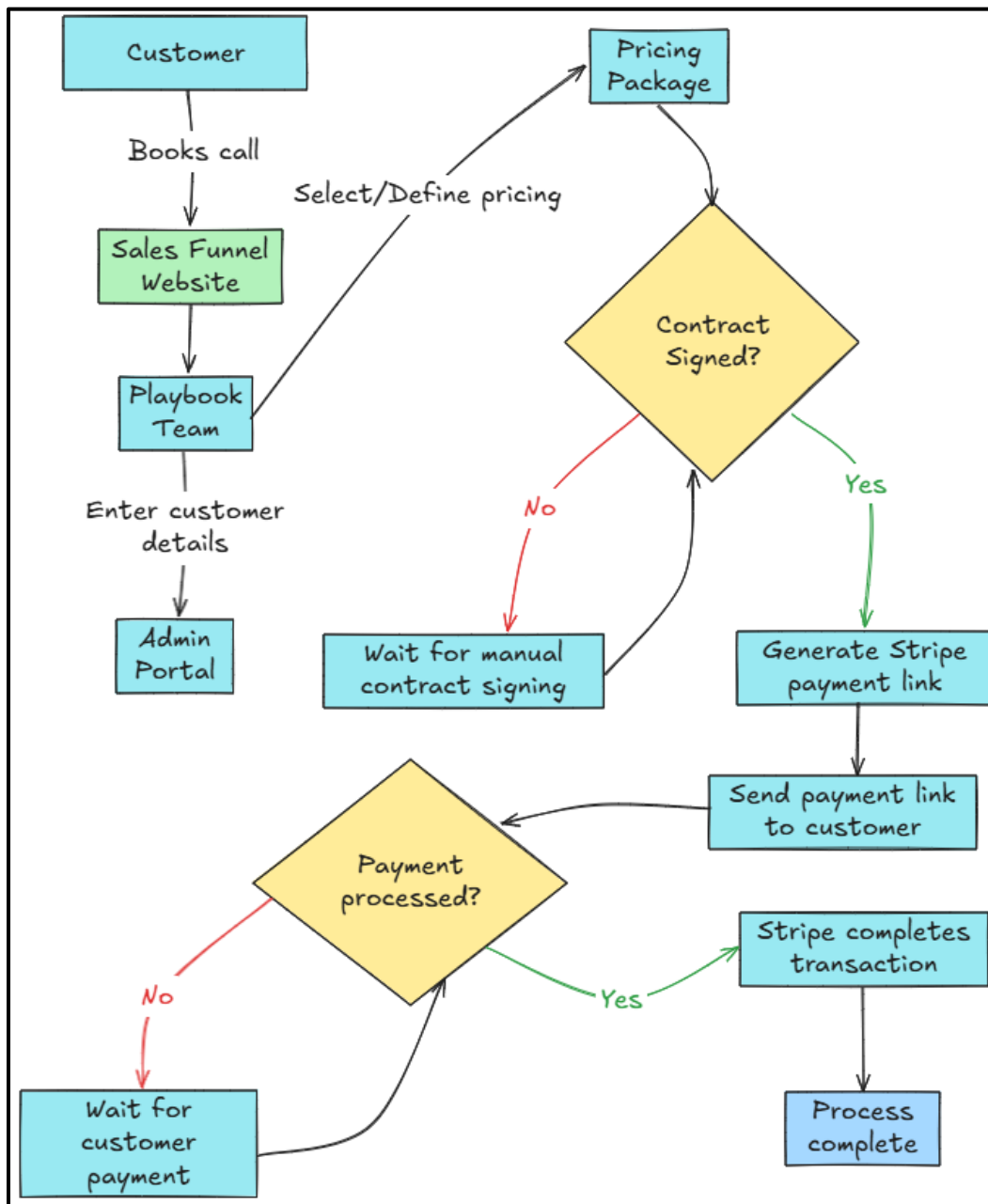
- a) Standard Credit Card Transactions:
 - Fee Structure: 1.5% + 0.2 GBP per transaction
 - Explanation
 - 1.5%: This percentage is deducted from the total transaction amount. For example, a 100 GBP transaction would incur a fee of 1.5 GBP.
 - 0.2 GBP: This is a fixed fee added to each transaction. So, the total fee for a 100 GBP transaction would be 1.5 GBP (percentage) + 0.2 GBP (fixed) = 1.7 GBP
- b) Premium Credit Card Transactions:
 - Fee Structure: 1.9% + 0.2 GBP per transaction
 - Explanation:
 - 1.9%: A higher percentage fee compared to standard cards, reflecting the additional benefits or rewards associated with premium cards.
 - 0.2 GBP: The fixed fee remains the same as with standard cards. Total Fee for 100 GBP Transaction: 1.9 GBP (percentage) + 0.2 GBP (fixed) = 2.1 GBP
- c) Understanding the Components of Stripe Fees:
 - Percentage Fee (%): This is a variable fee based on the transaction amount. It compensates Stripe for processing the payment and providing secure, reliable infrastructure. Higher transaction amounts naturally incur higher percentage fees.
 - Fixed Fee (Per Transaction): This is a flat fee charged for each transaction, regardless of the transaction amount. It covers the cost of handling each individual transaction, including aspects like authorisation, settlement, and fraud prevention.
- d) Example Scenarios: To illustrate how these fees work in practice, consider the following examples:
 - UK Standard Card Transaction:
 - Transaction Amount: 100 GBP
 - Fee Calculation: (1.5% of 100 GBP) + 0.2 GBP = 1.5 GBP + 0.2 GBP = 1.7 GBP
 - Net Amount Received: 100 GBP - 1.7 GBP = 98.3 GBP
 - UK Premium Card Transaction:
 - Transaction Amount: 100 GBP

- Fee Calculation: $(1.9\% \text{ of } 100 \text{ GBP}) + 0.2 \text{ GBP} = 1.9 \text{ GBP} + 0.2 \text{ GBP} = 2.1 \text{ GBP}$
- Net Amount Received: $100 \text{ GBP} - 2.1 \text{ GBP} = 97.9 \text{ GBP}$

15.5 Security and Compliance

- a) Data Security: Stripe's PCI DSS compliance and advanced security measures ensure that all payment data is handled securely. Tokenisation and encryption are employed to protect sensitive information during transmission and storage.
- b) Regulatory Compliance: Stripe assists canari in adhering to global financial regulations, including GDPR, CCPA, and local tax laws in the UAE, UK, and USA. Automated tax calculations and compliance reporting tools simplify regulatory adherence.
- c) Fraud Prevention: Stripe Radar's machine learning models detect and prevent fraudulent transactions, safeguarding both canari and its clients from financial fraud and chargebacks.

15.6 Workflow



To clarify the Stripe payment flow as outlined in the above diagram, steps:

- Customer Booking: Each customer manually books a call through the sales funnel website.
- Entering Customer Details: The canari team manually enters the customer's details into the admin portal.
- Selecting or Defining Pricing: The canari team selects an existing pricing package or defines a new price tailored to the specific customer.
- Sending Contract for Signing: A contract is sent to the customer for signing using their preferred tool, handled manually by the Canari team.

- e) Generating Stripe Payment Link: Once the contract is signed and the pricing is confirmed, a Stripe payment link is programmatically generated and sent to the customer.
- f) Completing Payment: The customer completes the payment through the Stripe link, which manages the transaction.

Note: There will be no dedicated payment pages on either the client or admin portals; all payments are facilitated exclusively through the Stripe link provided to the customer by the Canari team.

16. CMS – Webflow

Webflow is selected as the Content Management System (CMS) for the canari platform due to its robust features, ease of use, and seamless integration capabilities. As a powerful visual web design tool, Webflow enables the creation of dynamic, responsive websites without the need for extensive coding knowledge. Its CMS functionality allows for efficient management of content, making it an ideal choice for handling the diverse and growing content requirements of canari.

- a) Plan Details: Based on the platform's current and anticipated requirements, the CMS Plan (\$23/month billed yearly) emerges as the most suitable option. The key reasons for this selection include:
 - Bandwidth Requirements:
 - CMS Plan: 50 GB/month – Adequate for current traffic with room for growth.
 - Content Management Needs (CMS Collections and Items):
 - CMS Plan: 20 CMS collections and 2,000 CMS items – Ideal for managing blogs, FAQs, testimonials, and other dynamic content types.
 - Form Submissions:
 - CMS Plan: 1,000 form submissions/month – Sufficient for handling user interactions and feedback.
 - Additional Features:
 - Custom Domain: Essential for professional branding, available from the Basic Plan onward.
 - Surge Protection: Ensures website stability during traffic spikes.
 - Site Search and Localisation Preview: Enhances user experience and supports content customisation for different regions.
 - Editors: Up to 3 editors facilitate collaborative content management without incurring additional costs.
- b) Integration and Management: Webflow's CMS will be integrated into the admin portal of canari, allowing administrators to easily manage content, including the integration of third-party data such as holiday calendars. The admin panel will feature:
 - Data Entry Interface:
 - Intuitive forms for adding, editing, and deleting content within Webflow's CMS collections.
 - Options to import holiday data via API feeds or manual entry, supporting both automated and customisable data integration.
 - Customisation Options:

- Ability to specify regions, types of holidays, and relevant metadata to enhance analytics and content relevance.
- Localisation tools to tailor content for different audiences, supported by Webflow's localisation preview feature.
- Content Management:
 - Seamless handling of dynamic content such as blogs, FAQs, client testimonials, and more, leveraging Webflow's CMS collections and items.
 - Collaborative content editing facilitated by multiple editors, ensuring efficient and coordinated content updates.

17. Infrastructure Requirement

The SaaS platform is designed to support three primary components: the Client Portal, the Admin Portal, and the Website. The infrastructure leverages AWS and Azure as cloud service providers to meet the platform's scalability, reliability, and performance requirements. The Website is hosted using Webflow's CMS Plan, while the backend and frontend operations of the Client and Admin Portals are supported by AWS and Azure, depending on the chosen configuration. Below is the complete infrastructure breakdown, including metrics, specifications, and cost estimations for both cloud platforms.

17.1 Scenario

In this scenario, the platform is expected to accommodate an average of 100 users each day, with every user maintaining a single phone line. This usage pattern generates approximately 250 calls daily, each lasting around 2 minutes. Additionally, the platform will process 17 distinct analytics tasks to monitor performance, user engagement, and system health. This setup tests the infrastructure's ability to handle concurrent call operations and data analytics efficiently. The following are the platform components:

- a) Client Portal: Serves as the primary interface for clients to interact with the platform.
 - Metrics:
 - Page Size: 4 MB
 - Average Visitors Per Month: 3,000 (100 visitors/day)
 - Average Page Visits Per Visitor: 4/day
 - Daily Bandwidth: 1.5625 GB/day
 - Monthly Bandwidth: 46.875 GB/month
 - Maximum Request Size: 14 MB
 - Server Allocation:
 - 1 server for the frontend (user-facing elements)
 - 1 server for the backend (API calls and data processing)
 - Interactions: Handles approximately 11.5 million POST requests to the frontend per month, managed via AWS CloudFront or Azure CDN.
- b) Admin Portal: Provides tools for administrators to manage the platform and perform backend operations.
 - Metrics:
 - Page Size: 4 MB

- Average Visitors Per Month: 150 (5 visitors/day)
 - Average Page Visits Per Visitor: 5
 - Daily Bandwidth: 100 MB/day
 - Monthly Bandwidth: 3 GB/month
 - Server Allocation:
 - 1 server for the frontend (admin-facing UI)
 - 1 server for backend operations (business logic and workflows).
 - Interactions: Includes the same capabilities as the client portal, with additional backend processing for admin workflows.
- c) Website (Hosted on Webflow CMS): A public-facing marketing site to promote the platform and attract users.
- Metrics:
 - Page Size: 1 MB
 - Average Visitors Per Month: 15,000 (500 visitors/day)
 - Average Page Visits Per Visitor: 1
 - Daily Bandwidth: 500 MB/day
 - Monthly Bandwidth: 14.648 GB/month.
 - Hosting: Managed on Webflow's CMS Plan at \$23/month (billed yearly).
 - Webflow Features:
 - Bandwidth: 50 GB/month
 - CMS Functionality: 20 CMS collections, 2,000 CMS items
 - Form Submissions: 1,000 per month
 - Additional Features: Custom domain support, surge protection, and SEO tools
- d) AI Model: Provides AI-driven analytics and insights for the platform, requiring high GPU performance.
- Metrics:
 - Runtime: 24,000 minutes/month (400 hours)
 - GPU Memory Requirements: 16 GB GPU RAM
- e) AWS Cost Breakdown

Service	Monthly Cost	Details
Amazon EC2	\$25.19	2 t3a.medium VMs for client and admin portals
Amazon RDS	\$343.52	PostgreSQL database (40 GB SSD, Multi-AZ)
Amazon CloudFront	\$2,672.19	CDN for frontend delivery and caching
AI Model Hosting	\$300.80	EC2 GPU instance (g4dn.2xlarge)
Webflow CMS Plan	\$23.00	Hosting for the website
Total (AWS)	\$3,364.70.	Combined monthly cost for AWS and Webflow

f) Azure Cost Breakdown

Service	Monthly Cost	Details
Azure Virtual	\$380.00	4 VMs (D2s v3) for client and admin portals
Azure SQL Database	\$150.00	Standard S3 tier (100 DTUs, 250 GB)
Azure CDN	\$208.46	CDN for frontend delivery and caching

AI Model Hosting	\$472.00	NCas_T4_v3 GPU instance (16 GB VRAM)
Webflow CMS Plan	\$23.00	Hosting for the website
Total (Azure)	\$1,233.46.	Combined monthly cost for Azure and Webflow

g) Bandwidth Summary

Component	Daily Bandwidth	Monthly Bandwidth	Notes
Client Portal	1.5625 GB	46.875 GB	Handles user-facing interactions
Admin Portal	100 MB	3 GB	Includes backend workflows and admin UI
Website	500 MB	14.648 GB	Hosted on Webflow CMS

h) Web Form Setup

Component	Server Allocation
Component	Server Allocation
Website	Hosted on Webflow CMS Plan (\$23/month).
Client Portal	1 server for frontend, 1 server for backend (AWS EC2 or Azure VM)
Admin Portal	1 server for frontend, 1 server for backend (AWS EC2 or Azure VM)

i) Comparison between AWS and Azure

Category	AWS Cost	Azure Cost	Notes
Compute	\$25.19	\$380.00	Azure includes 4 VMs, AWS uses 2 VMs and CloudFront
Database	\$343.52	\$150.00	Azure SQL is more cost-effective than Amazon RDS
CDN	\$2,672.19	\$208.46	AWS CloudFront is costlier; Azure CDN offers similar service
AI Hosting	\$300.80	\$472.00	GPU costs for 16 GB VRAM are higher on Azure
Webflow CMS Plan	\$23.00	\$23.00	Website hosting cost remains the same
Total	\$3,364.70	\$1,233.46	Azure remains more cost-effective overall

j) Conclusion:

- AWS Infrastructure: AWS provides a scalable and comprehensive solution but comes with a higher total cost of \$3,364.70/month due to higher CDN and database costs.
- Azure Infrastructure: Azure offers significant cost savings, with a total monthly cost of \$1,233.46/month, despite slightly higher GPU costs for the AI model.